



BULACAN STATE UNIVERSITY



COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

City of Malolos, Bulacan

IT 312 Analytics Tools and Techniques

Project Requirements

2nd Semester 2024-2025

Project Title		Safer Calumpit: Predicting Local Crime Through Analysis of Blotter Records			
Team Members	Arrange your names based on the percentage of members' contribution. Student 1 with the most contribution.		Score		
	Last name, First name	Brief Description of Members' Contribution	Individual Presentation (100)	Group Grade	
				Data Pre-Processing	Insight Presentation Using Dashboard
1	Chico, Ken	Data Collector, Data Preprocessor, Analyst, Documentation, System Developer, Dashboard Insights, Live Presentor, Quality Checker			
2	De Jesus, Princess Anne	Data Collector, Data Preprocessor, Analyst, Documentation, UI/UX Designer, Dashboard Designer and Insights, Video Specialist			
3	Garces, Grace	Data Collector, Data Preprocessor, Analyst, Documentation, System Developer, Dashboard Insights, Video Specialist			
4	Lachica, Crystel	Data Collector, Data Preprocessor, Analyst, Documentation, System Developer, Dashboard Insights, Video Specialist			
5	Macabenta, Laarnie	Data Collector, Data Preprocessor, Analyst, Documentation, UI/UX Designer, Dashboard Designer and Insights, Video Specialist			
6	Mendoza, Carl Andrei	Data Collector, Data Preprocessor, Analyst, Documentation, UI/UX Designer, Dashboard Designer and Insights, Live Presentor, Project Leader			
Total Group Grade:					



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IT 312: ANALYTICS TOOLS AND TECHNIQUES

Safer Calumpit: Predicting Local Crime Through Analysis of Blotter Records

Project in IT312

Submitted by:

Chico, Ken D.L.

De Jesus, Princess Anne D.G.

Garces, Grace S.

Lachica, Crystel H.

Macabenta, Laarnie R.

Mendoza, Carl Andrei B.

Submitted to:

Dr. Rosemarie M. Bautista

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Introduction

a. Project Rationale

A crime is a deliberate act that causes physical or psychological harm, damage to, or loss of property, and is punishable by law. Crime threatens the peace and safety of the people, making crime analysis crucial. With the advent of data mining and machine learning, traditional and manual crime analysis proves to be outdated (Saeed & Abdulmohsin, 2023). By utilizing these, the analysis would help law enforcement agencies in using resources effectively, anticipate crime at a specific time, and protect the safety and well-being of the people (Khan et al., 2022).

This project seeks to assist Calumpit Municipal Police Station (CMPS) of Bulacan, under the umbrella of the Philippine National Police (PNP) in analyzing crime-related data to identify patterns, enhance response strategies, and inform decision-making processes by analyzing crime rates over time, identifying peak crime periods, and understanding the distribution of different types of crimes across various locations within Calumpit and this project will provide valuable insights through visualizations which includes charts, graphs, and maps that highlight trends and patterns in crime data. The project will help CMPS, more specifically, the head investigator and their team, in analyzing crime patterns and incident management within the confines of Calumpit, thereby contributing to the overall safety and well-being of the community.

b. Objectives of the Study

This project aims to assist law enforcement agencies in analyzing crime-related data to identify patterns, enhance response strategies, and inform decision-making processes. It focuses on improving the accuracy of law enforcement activities through data-driven insights.

1. Enhance community safety through data-driven strategies by analyzing historical crime trends in Calumpit.
2. Reduce the frequency of crime incidents in identified hotspot areas or high-risk areas
3. Explore relationships between crime types and external factors such as location, or victim demographics.

Analytics Framework

Figure 1

Analytics Framework of Objective 1

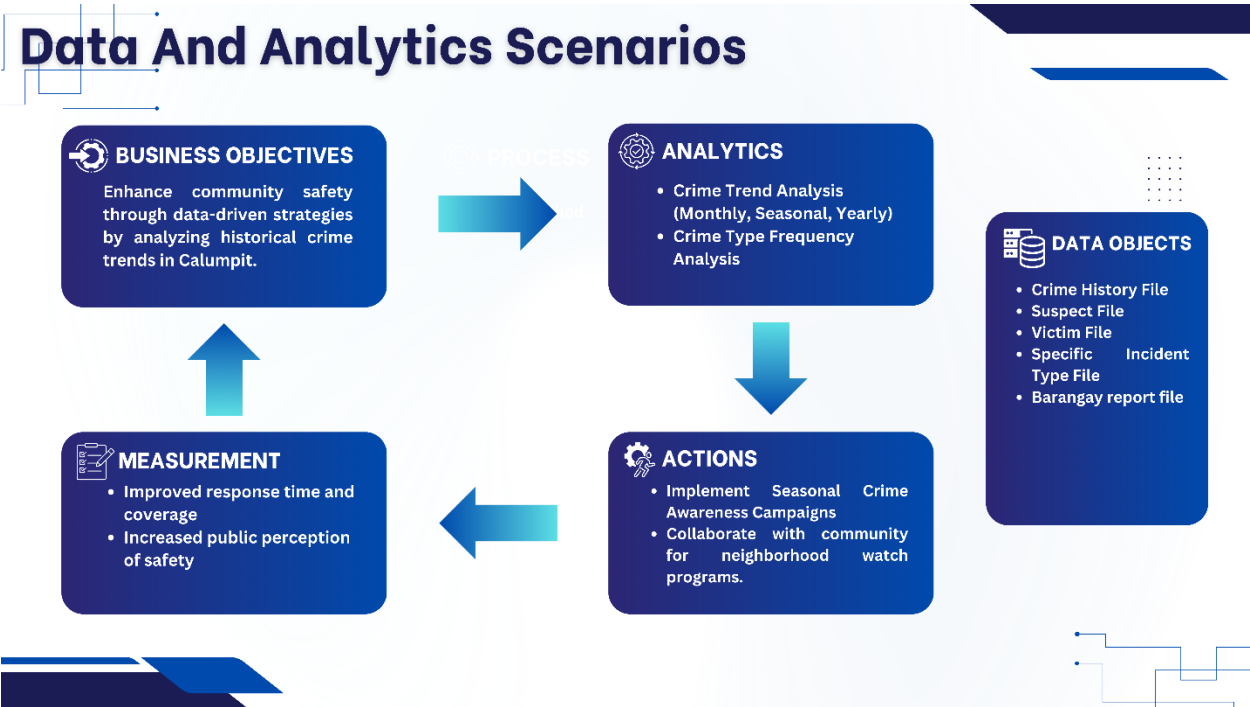


Figure 2

Analytics Framework of Objective 2

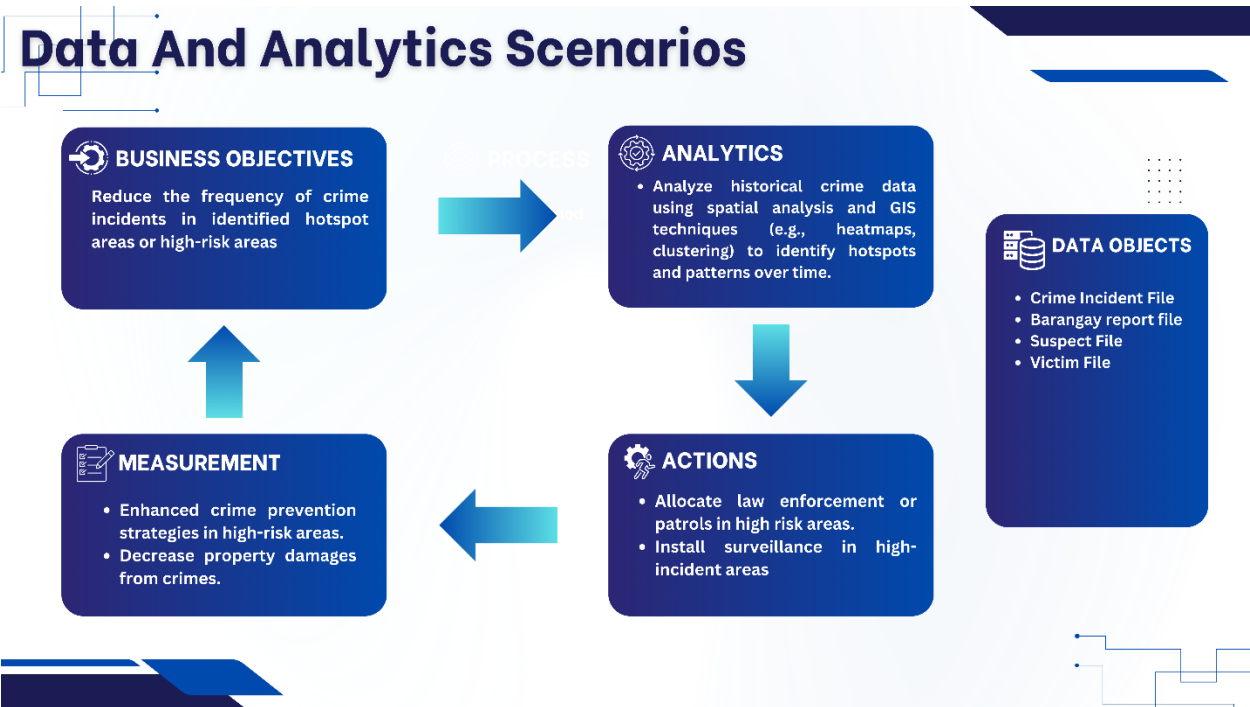
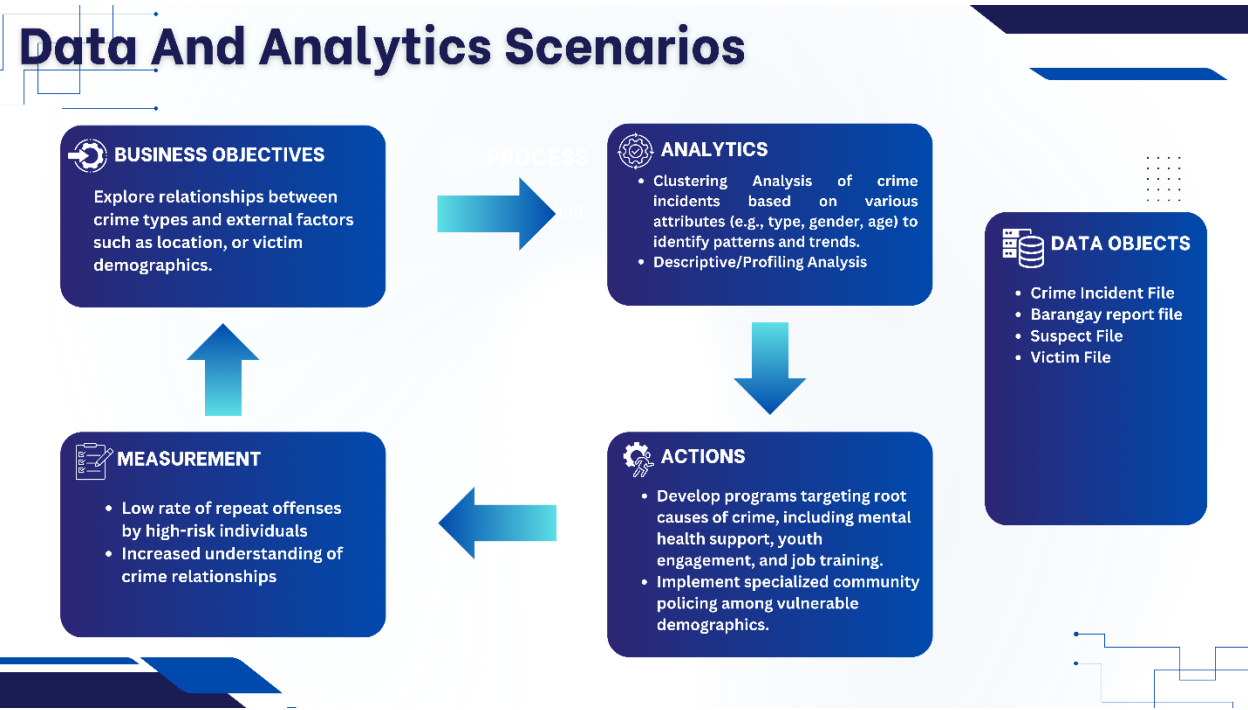


Figure 3

Analytics Framework of Objective 3



Dataset Overview

a. Original Dataset

A	B	C	D	E	F	G	H	I	J
blotterno	dateEncoded	pro	ppo	stn	pcp	region	province	municipal	barangay
031407-202201-2014-01-0001	2022-01-12 14:16:08	PRO 3	BULACAN PPO	Calumpit MPS		CENTRAL LUZON	BULACAN	CALUMPIT	IBA O'ESTE
031407-202201-2014-01-0019	2022-01-12 14:43:34	PRO 3	BULACAN PPO	Calumpit MPS		CENTRAL LUZON	BULACAN	CALUMPIT	PALIMBANG
031407-202204-2013-04-38W	2022-04-18 11:02:13	PRO 3	BULACAN PPO	Calumpit MPS		CENTRAL LUZON	BULACAN	CALUMPIT	SAN MIGUEL
031407-202208-W-2013-04-0383	2022-08-22 09:43:33	PRO 3	BULACAN PPO	Calumpit MPS		CENTRAL LUZON	BULACAN	CALUMPIT	CALIZON

K	L	M	N	O	P	Q	R	S
street typeofPlace	dateReported	timeReported	dateCommitted	timeCommitted	incidentType	iscime	stageoffelony	
	2014-01-01	01:36:00	2013-01-01	00:36:00	(Incident) Vehicular Accident	YES	CONSUMMATED	
	2013-01-06	01:30:00	2013-01-05	00:57:00	(Incident) Vehicular Accident	YES	CONSUMMATED	
Residential (house/condo)	2013-01-18	08:45:00	2013-01-18	08:45:00	(Incident) Swindling/Estafa	YES	CONSUMMATED	
Along the street	2013-04-21	19:15:00	2013-01-21	19:15:00	(Incident) ERI - Destruction Of Property	YES	CONSUMMATED	

T	U	V	W	X
offense	offenseType	section	modus	victim
RECKLESS IMPRUDENCE RESULTING TO PHYSICAL INJURY - RPC Art 365	Crimes Against Persons			Glenn Carlo Nabong Ramos (12/Male/Injured/FILIPINO/)
RECKLESS IMPRUDENCE RESULTING TO DAMAGE TO PROPERTY - RPC Art 365	Crimes Against Property			RICARDO PANTUA CAPINA (44/Male/Unharmed/FILIPINO/)
SWINDLING (ESTAFA) - RPC Art. 315 as amended by PD 1689	Crimes Against Property			jun Unknown motalvo (41/Female/Unharmed/FILIPINO/)
RECKLESS IMPRUDENCE RESULTING TO DAMAGE TO PROPERTY - RPC Art 365	Crimes Against Property			VICTOR DONDONILLA DE JESUS (55/Male/Released/FILIPINO/)

Y	Z	AA	AB	AC	AD	AE	AF	AG
suspect	threatGrp	grpAffiliation	incidenttypethreatgrp	mrs	suspectisEGO	suspectEGOPosition	suspectEGOClass	victimisEGO
NIÑO NELSON Unidentified YANGA (38/Male/Released (No Complainant)/FILIPINO/)	No				No			No
	No				No			No
ANALYN Unidentified VALDERAMA (32/Female/Released/FILIPINO/)	No				No			No
CARLOS UY DIONISIO (51/Male/At-Large/FILIPINO/)	No				No			No

AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ
victimEGOPosition	victimEGOClass	suspectMotive	suspectSubMotive	victimKilled	victimInjured	victimUnharmed	suspectKilled	suspectAtLarge	suspectArrested
				No	Yes	No	No	No	No
				No	No	Yes	No	No	No
				No	No	Yes	No	No	No
				No	No	No	No	Yes	No

Safer Calumpit: Predicting Local Crime Through Analysis of Blotter Records

AR		AS	AT		AU	AV	AW	AX
narrative		casestatus	caseSolveType	caseStatusinProsec	groundDismissedProsec	fillingType	dateFiledtoProsec	
DTPO On January 01 2013 at around 01 00 o clock in the morning a vehicular accident was transp		Solved	SOLVED (AMICABLY SETTLED)					
DTPO On January 5 2013 at around 12 30 o clock in the morning a vehicular accident was transpir		Solved	SOLVED (AMICABLY SETTLED)					
On this time and date the certaion person Marilou Montalvo y Borlongan 39 y o married resident		Solved	SOLVED (AMICABLY SETTLED)					
RIR TO DAMAGED TO PROPERTY INITIAL INVESTIGATION REVEALED THATBOTH VEHICLES TRAVE		Cleared						

AY	AZ	BA	BB	BC	BD	BE	BF	
ISDocketNo	prosecutor	caseCourtStatus	dateFiledInCourt	ccNo	judge	court	groundDismissedCourt	
NPS Nr III-04-INC Asst.Prov.Pro	Pros. Avelino Basco	DECIDED - DISMISSAL	2016-12-07	420-M-2014	Victoria Fernandez-Ber	RTC BR 18	FAILURE OF LAW ENFORCEMENT TO ATTEND COURT HEARINGS	
NPS Nr III-04-INC Asst.Prov.Pro	Pros. Avelino Basco	DECIDED - DISMISSAL	2016-12-07	420-M-2014	Victoria Fernandez-Ber	RTC BR 18	FAILURE OF LAW ENFORCEMENT TO ATTEND COURT HEARINGS	
NPS Nr. III-04-IN Asst.Prov.Pro	Pros. Fredierick Malapit.							
NPS Nr. III-04-IN Asst.Prov.Pro	Pros. Fredierick Mal	ON-GOING TRIAL	2014-02-25	810-M-2014	JUDGE MONTERO JR.	RTC BR 8		

BG	BH	BI	BJ	BK		BL	BM	BN	BO	BP	BQ
lat	lng	victimCount	suspectCount	investigator		headInves	placeCommission	robStabishmentType	robStabishmentName	faserialno	facaliber
14.912582	120.75206	1	1	SPO3 Rommel Manansala - 09985985380		SPO3 Rommel Manansala - 09985985380					
14.885646	120.782761	1	1	SPO3 Rommel Manansala - 09985985380		SPO3 Rommel Manansala - 09985985380					
14.889544	120.783089	1	1	PO2 Ferdie Santos - 09985985380		SPO3 Rommel Manansala - 09985985380					
14.900027	120.776222	1	1	PO3 Cherry Rodriguez - 09985895380		SPO3 Rommel Manansala - 09985895380					

BR	BS	BT		BU	BV	BW	BX	BY
fakind	famake	fastatus		drugInvolved	vehiclePlateNo	vehicleMCPlateNc	vehicleChassisNo	vehicleKind
					.RT 7944			TRUCK, MOTORCYCLE
					TGI 818	Fpi60683	LHI6CHH00AH077667	CAR, CAR
						150FMG13HII690		MOTORCYCLE

BZ	CA	CB	CC	CD	CE
vehicleMake	vehicleModel	vehicleStatus	vehicleRegistrationNo	vehicleRegistrationStatus	vehicleColorum
YAMAHA					No
JEEP, TOYOTA	FUSO JITNEY, COROLA				No

- b. Dataset Information
- **Title:** Safer Calumpit: Predicting Local Crime Through Analysis of Blotter Records
 - **Domain or Sector:** The dataset pertains to the law enforcement sector, specifically focusing on crime-related incidents and police procedures.
 - **Number of Records and Attributes:** The raw dataset contains 7950 records and 83 attributes that provide detailed information on various law enforcement activities.
 - **Time Period Covered by the Dataset:** The dataset spans from 2013 to 2025, capturing a comprehensive period of law enforcement data.
 - **Credibility and Relevance of the Data Source:** The data is credible and relevant, as it is based on actual procedures and workflows followed by the Calumpit PNP (Philippine National Police), ensuring accuracy and reliability in the representation of law enforcement activities.

Attribute Profile Table

Attribute Name	Data Type	Description	Possible Values / Range
barangay	String	A unit of local government, typically a village or district, within a city or municipality in the Philippines.	Balite, Balungao, Buguion
typeofPlace	String	The type of location where the incident took place.	Residential, Commercial,
dateComitted	Date	The date on which the incident occurred.	2025-05-02
timeComitted	Time	The specific time when the incident occurred.	12:36AM
incidentType	String	The general category of the incident.	Incident, Operation
incidentTypeSpecific	String	A more detailed description of the crime.	Kidnapping, Libel
stageOffelony	String	The current stage or progress of the offense.	Frustrated, Attempted
offenseType	String	The legal classification of the crime.	Acts Against Property Only, Injurious Acts of a Sexual Nature
Modus	String	The method or pattern used by the suspect to commit the crime.	Extortion, Hacking
VictimAge	Integer	The age of the victim at the time of the incident.	00-14 (Children), 15-24 (Youth), 25-64 (Adult), 65+ (Senior)
VictimGender	String	The gender of the victim.	Male, Female
VictimSeverity	String	The degree of injury or harm the victim sustained.	Alive, Deceased
VictimOccupation	String	The job or profession of the victim.	Manager, Professionals
SuspectAge	Integer	The age of the suspect at the time of the incident.	00-14 (Children), 15-24 (Youth), 25-64 (Adult), 65+ (Senior)
SuspectGender	String	The gender of the suspect.	Male, Female
SuspectDisposition	String	The current legal status or condition of the suspect.	Custody Status, Escaped

SuspectOccupation	String	The job or profession of the suspect.	Manager, Professionals
victimKilled	Boolean	Whether the victim was killed in the incident.	Yes, No
victimInjured	Boolean	Whether the victim was injured in the incident.	Yes, No
victimUnharmred	Boolean	Whether the victim was unharmred in the incident.	Yes, No
suspectKilled	Boolean	Whether the suspect was killed in the incident.	Yes, No
suspectAtLarge	Boolean	Whether the suspect is currently at large.	Yes, No
suspectArrested	Boolean	Whether the suspect has been arrested.	Yes, No
Casestatus	String	The current status of the case.	Solved, Cleared
robStablishmentType	String	The type of establishment involved in an incident.	Education, Private Household
vehicleKind	String	The type of vehicle involved.	Bus, Car
vehicleColurum	Boolean	Indicates whether a vehicle is operating without proper registration or permit.	Yes, No

Attribute Value Discussion

Attribute Name	Why is this attribute important?	How it may contribute to your analysis	How it may be grouped or categorized (e.g., converting grades into pass/fail)
barangay	This attribute pinpoints the exact location within the Calumpit where the accident occurred.	Help to identify which locations have higher crime rates or are more prone to certain incidents.	The value in dataset is not applicable for categorization
dateCommitted	Allows the analysis of	Use as a timeline filter or for time-series graphs to	The values were extracted to get the

	historical trends and patterns using the date when the incident occurred.	show incident trends over time.	Day Name and Month Name.
timePeriod	Derived from the time committed variable, it simplifies and provides a way to analyze the time of day of incidents.	Shows incident distribution across time periods (e.g, Morning, Evening) for time-based crime trend analysis.	The value in dataset is not applicable for categorization
DayCommitted	Derived from the date committed variable, allows easy analysis of days for each occurrence of incidents.	Displays day-of-month incident patterns, detetcting specific days with higher crime occurrences.	The value in dataset is not applicable for categorization
MonthCommitted	Derived from the date committed variable, allows easy analysis of months for each occurrence of incidents.	Visualizes incident frequency by month, identifying seasonal or monthly trends in crime occurrence.	The value in dataset is not applicable for categorization
VictimGender	Allows the analysis of how gender can affect certain incidents.	This attribute will contribute to detecting the total number of incidents in gender based.	The value in dataset is not applicable for categorization
VictimInjured	Help in analyzing the frequency of victims getting injured against a particular incident.	Shows whether the victim was injured, contributing to the analysis of incident outcomes.	The value in dataset is not applicable for categorization

VictimKilled	Aids in finding out the most severe incidents that would lead to a victim's death.	Shows counts of incidents resulting in death, contributing to outcome-based severity analysis in the dashboard.	The value in dataset is not applicable for categorization
VictimSeverity	Indicates the status of the victim after an incident, helping in understanding the consequences of each incident.	Provides an overall view of incident consequences by showing the ratio of unharmed, injured, and deceased victims.	The value in dataset is not applicable for categorization
victimUnharmed	Indicating if the victim is unharmed helps in understanding incidents that are less severe compared to other incidents.	Highlights cases where victims were unharmed, balancing the overall view of incident outcomes.	The value in dataset is not applicable for categorization
victimAge	This attribute aids in analyzing the relevance of different age groups to find out the relationship between age groups and different crimes.	Enables age-distribution charts or filters, revealing which age groups are most affected by incidents.	Classified into standard age binning according to WHO this includes 0–14 years, 15–24 years, 25–64 years, and 65 and over.
VictimOccupation_Categorized	Provides context about the social or economic background of victims.	Helps to analyze if certain occupations are targeted in specific incidents.	Categorized into Manager, Professionals, Technicians and Associate Professionals, Clerical Support Workers, Service and Sales Workers,

			Skilled Agriculture, Forestry and Fishery Workers, Craft and Related Trades Workers, Plant and Machine Operators and Assemblers, Elementary Occupations, Armed Forces Occupations, and Unclassified.
casestatus	Allows further analysis between solved and unresolved cases and the type of incidents they typically fall to.	Indicates whether cases are solved or unresolved, helping assess case resolution performance	The value in dataset is not applicable for categorization.
incidentType	Identifies the general type of crime or event that occurred.	Helps categorize and quantify the most common types of incidents.	The value in dataset is not applicable for categorization.
offenseType	Specifies the legal classification of the offense.	Allows legal analysis and alignment with penal codes for reporting or judicial tracking.	Categorized based on the nature and severity of the criminal acts they represent. These classifications draw upon government documentation, research data, and established standards to provide a consistent and meaningful way to group
stageoffelony	Provides deeper meaning about the progress of the crime and	Highlights cases where victims were unharmed, balancing the overall	The value in dataset is not applicable for categorization

	how it may relate to each incident occurrence.	view of incident outcomes.	
SuspectAge	Helps in understanding what age group commits different kinds of incidents and crimes.	Shows demographic patterns among suspects, supporting offender profiling through age-distribution visualizations.	Classified into standard age binning according to WHO this includes 0–14 years, 15–24 years, 25–64 years, and 65 and over.
SuspectGender	Allows the analysis of how gender can influence the occurrence of certain crimes.	Visualizes the gender distribution of suspects, helping detect crime patterns related to gender.	The value in dataset is not applicable for categorization
SuspectKilled	Provides the analysis of how certain crimes can lead to an offender’s death.	Shows the number of suspects killed during incidents, providing insights for operational and legal review.	The value in dataset is not applicable for categorization
suspectArrested	Helps in understanding which type of crimes suspects are more likely to be arrested or not.	Shows how frequently suspects are apprehended, helping measure law enforcement response and incident resolution.	The value in dataset is not applicable for categorization
SuspectAtLarge	Finds the types of crimes of which suspects are more likely to be in custody or escape from.	Displays ongoing cases with suspects at large, contributing to operational alerts and case follow-up priorities.	The value in dataset is not applicable for categorization
SuspectDisposition_Categorized	Gives details on what happened to the suspect after the incident.	Helps determine law enforcement effectiveness and outcomes in the side of	Categorized into legal resolution status (acquitted, released, probation, serving sentence),

		the suspect for each case.	custody status (detained, arrested, released, turned over), physical condition (hospitalized, wounded, deceased), escape status, and voluntary surrender.
SuspectOccupation_Categorized	It helps to better understand the livelihood of the suspects which may be related to their crime(s).	May reveal patterns related to socio-economic status and crime tendencies.	Categorized into Manager, Professionals, Technicians and Associate Professionals, Clerical Support Workers, Service and Sales Workers, Skilled Agriculture, Forestry and Fishery Workers, Craft and Related Trades Workers, Plant and Machine Operators and Assemblers, Elementary Occupations, Armed Forces Occupations, and Unclassified
typeofPlace_Categorized	Simplified and categorized the values of type of place for uniformity.	It will identify what environments are riskier and will contribute to allocating more officers in that area.	Grouped into several categories based on their characteristics and functions, including residential areas (abandoned structures and occupied houses), utilities and infrastructure (street locations and

			transportation terminals), parks and recreation (natural spaces and recreational places), protected areas (rivers and lakes), and institutional categories (cemeteries, government offices, hospitals, religious places, and educational institutions).
incidentTypeSpecific_Categorized	It is the categorized values of specific types of incidents derived from the narrative of the witness.	Provides detailed insights on specific incident types, allowing users to drill down into categories for trend analysis and decision-making support.	Clustered specific type of incident based on similar characteristics such as (Vehicular Incident, Physical Assault)
robStablishmentType_Categorized	Provides a more simplified and uniform description of the type of establishment that was robbed of a more precise analysis.	Displays incidents by establishment type (e.g., bank, store), helping detect vulnerable business types.	Classified into Retail & Wholesale Trade stores, educational institutions (colleges, schools, universities), and Private Households (residential).
VictimAge_Categorized	Specifying the age group for the victim age makes the analysis easier and more interpretable.	This helps in recognizing vulnerable populations — for example, if incidents affecting children or seniors are increasing, it could prompt targeted safety measures, community education programs, or	Classified into children (0–14 years), youth (15–24 years), adults (25–64 years), and seniors (65 and over).

		protective policies for those groups.	
SuspectAge_Categorized	Specifying the age group for the suspect age makes the analysis easier and more interpretable.	It enables analysis of crime tendencies by age category — like whether youth or adults are more often involved in types of incidents — which can help in designing age-targeted crime prevention programs and interventions.	Classified into children (0–14 years), youth (15–24 years), adults (25–64 years), and seniors (65 and over).
Victim Profile	Summarized and simplified values of Victim Age, Gender, and Barangay used for certain visuals in the dashboard.	This enables pattern detection of who the common victims are and where incidents involving them usually occur, supporting focused community safety planning, policy development, and preventive interventions.	Grouped victims by age, gender ("Male"), and barangay location.
Victim ProfileCategorized	Summarized and simplified values of Victim Age Group derived from Victim Age, Gender, and Barangay used for certain visuals in the dashboard.	This allows the analysis to identify which specific demographic profiles are most frequently affected in certain barangays, helping prioritize safety measures, resource allocation, and targeted awareness campaigns for high-risk groups.	Grouped victims by age category ("Adult"), gender ("Male"), and barangay location.

Preprocessing Techniques

OPEN REFINE

Figure 4

Split Multivalued Victims and Suspect

OpenRefine

BSU 2013 2025.xlsx

Permalink

Open...

Export

Help

Facet / Filter

Undo / Redo 0 / 0

6,082 rows

Extensions Wikibase

Show as: rows records

Show: 5 10 25 50 100 500 1000 rows

« first

< previous

1

- 10

next >

last »

modus	victim	suspect	three	group
	Glenn Carlo Nabong Ramos (12/Male/Injured/FILIPINO/)	NIÑO NELSON Unidentified YANGA (38/Male/Released (No Complainant)/FILIPINO/)	No	
	RICARDO PANTUA CAPINA (44/Male/Unharmed/FILIPINO/)		edit No	

OpenRefine

BSU 2013 2025.xlsx

Permalink

Open...

Export

Help

Facet / Filter

Undo / Redo 0 / 0

6,082 rows

Extensions Wikibase

Show as: rows records

Show: 5 10 25 50 100 500 1000 rows

« first

< previous

1

- 10

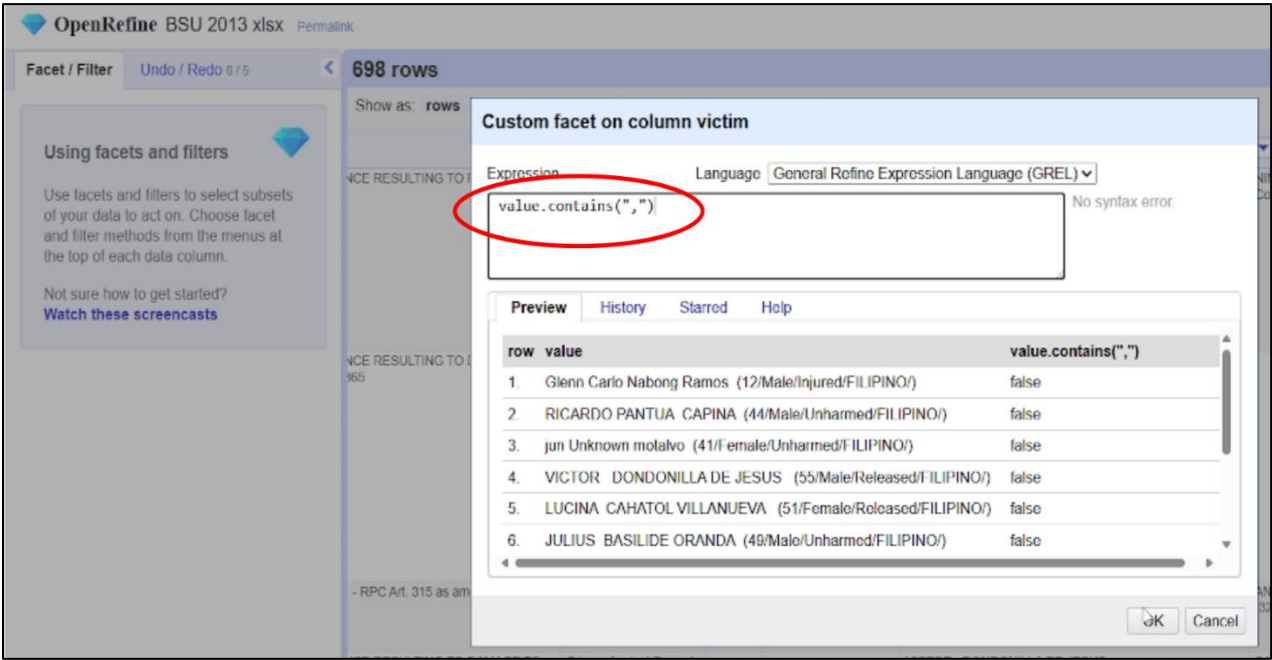
next >

last »

modus	victim	suspect	three	group
	JULIUS BASILIDE ORANDA (49/Male/Unharmed/FILIPINO/)	JOCELYN MABINA CUAJAO (49/Male/At-Large/FILIPINO/)	No	
	BENJIE DETAL CORTEZ (26/Male/Injured/FILIPINO/), ERIC DETAL CORTEZ (25/Male/Injured/FILIPINO/)	JOEL SALONGA UMALI (43/Male/Released (No Complainant)/FILIPINO/)	No	
Hitting with hand object	BELEN SISCA MARSO (29/Female/Injured/FILIPINO/)	CHRISTOPHER CRUZ TAROSA (37/Male/Released/FILIPINO/NONE)	No	
	BELEN SISCA MARSO (59/Male/Injured/FILIPINO/)	CHRISTOPHER TAROSA CRUZ (34/Male/Released (No Complainant)/FILIPINO/)	No	

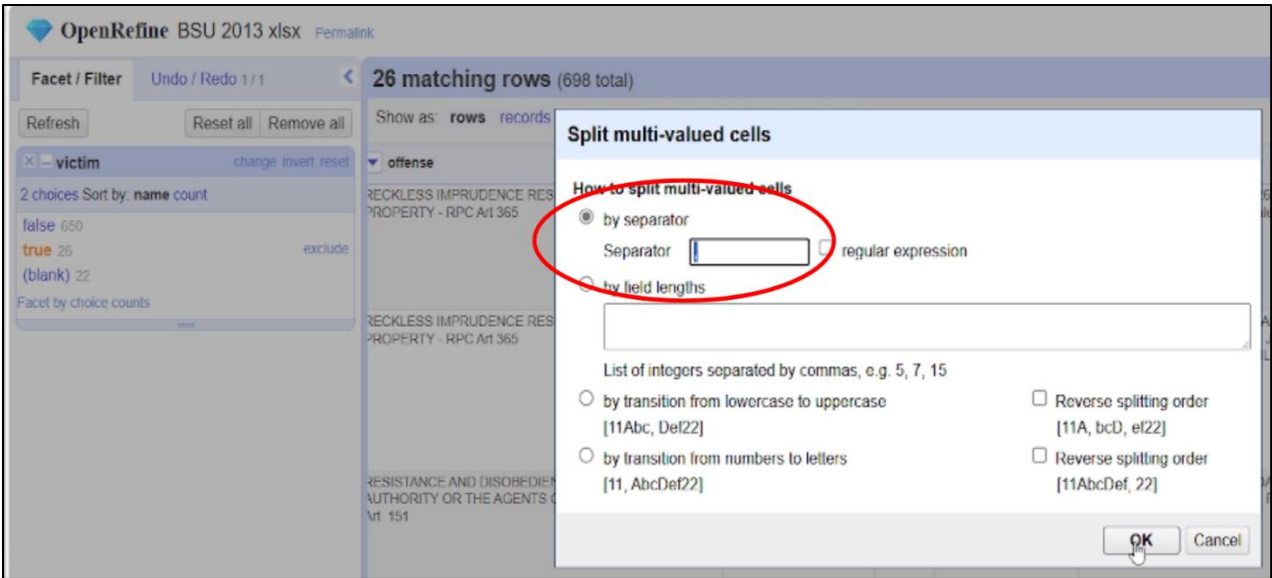
Splitting records that contain multiple instances in either the victim or suspect columns into individual rows for each entry.

Figure 5
Custom Facet to Search for Multivalued Cells



This figure shows how to split the information of a victim and suspect using conditional formatting. The raw dataset used comma (",") to input multiple people in a single cell. By creating a custom facet, it will only display multivalued cells making it easy to check and split multivalued attributes.

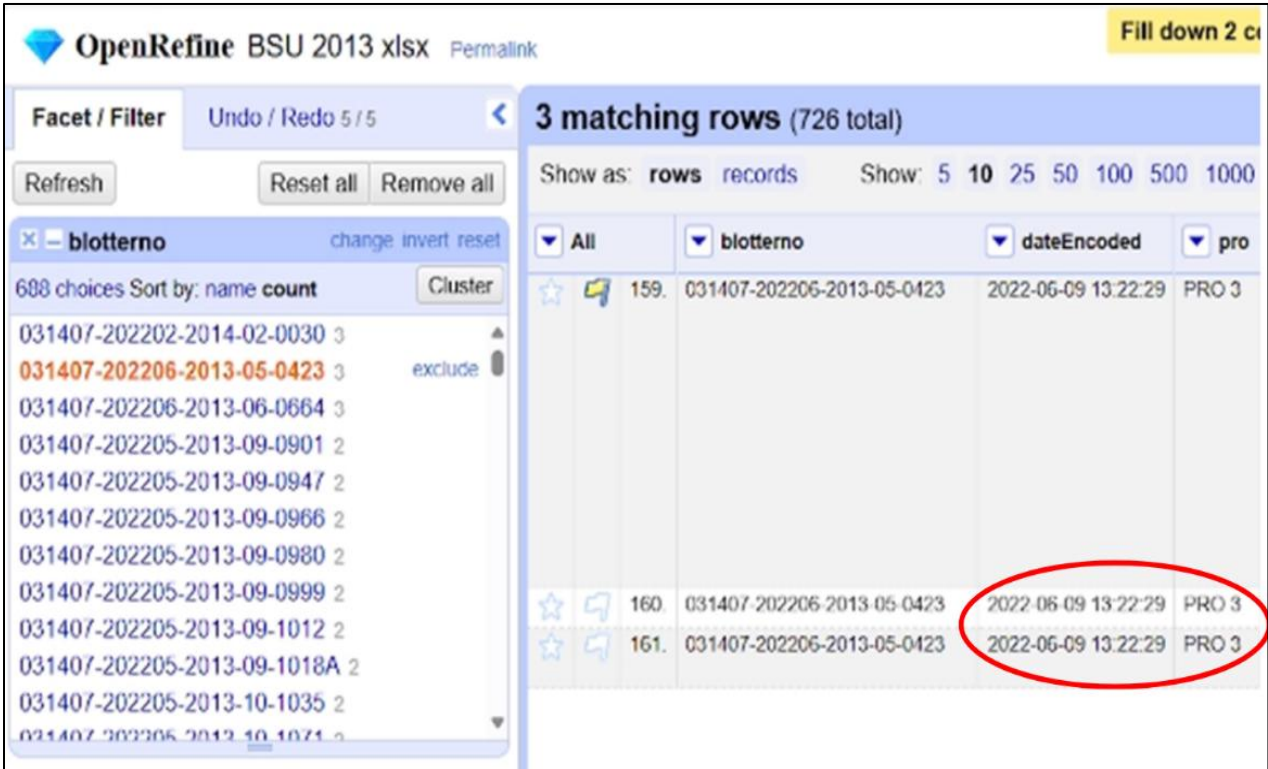
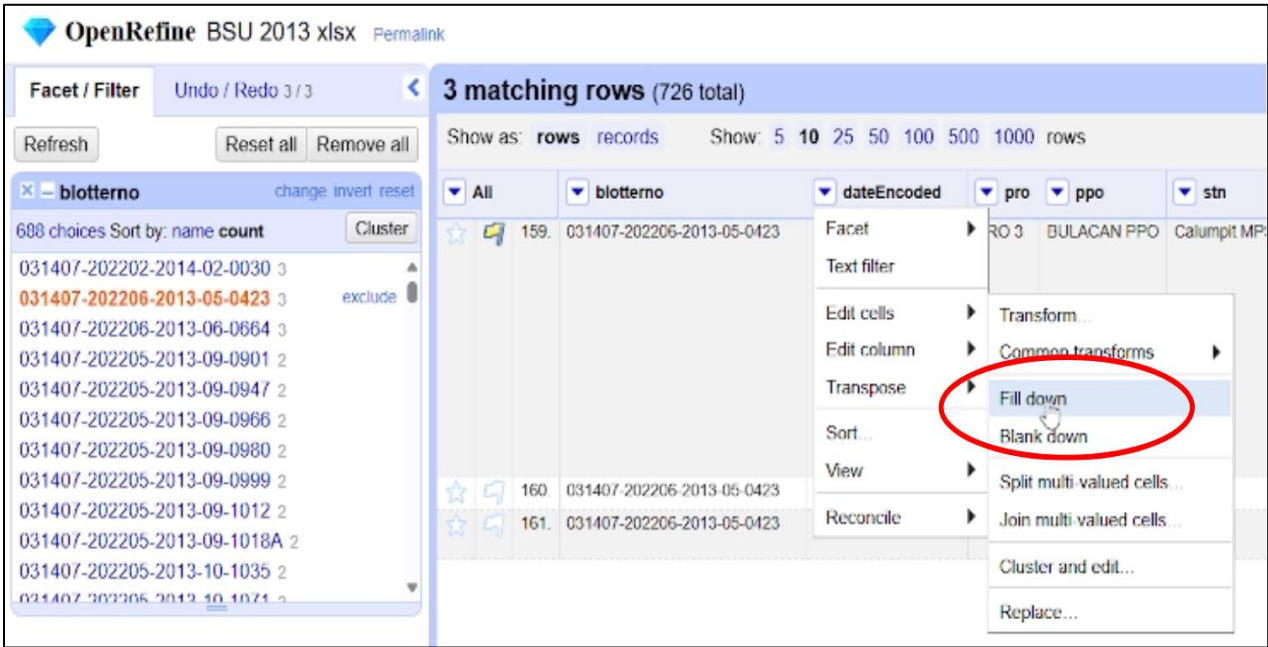
Figure 6
Split the Text Based on the Separator



This figure shows how we split the text based on a separator. The operator used is a comma (",") for splitting the text into distinct parts.

Figure 7

Fill Down with Same Blotter No.

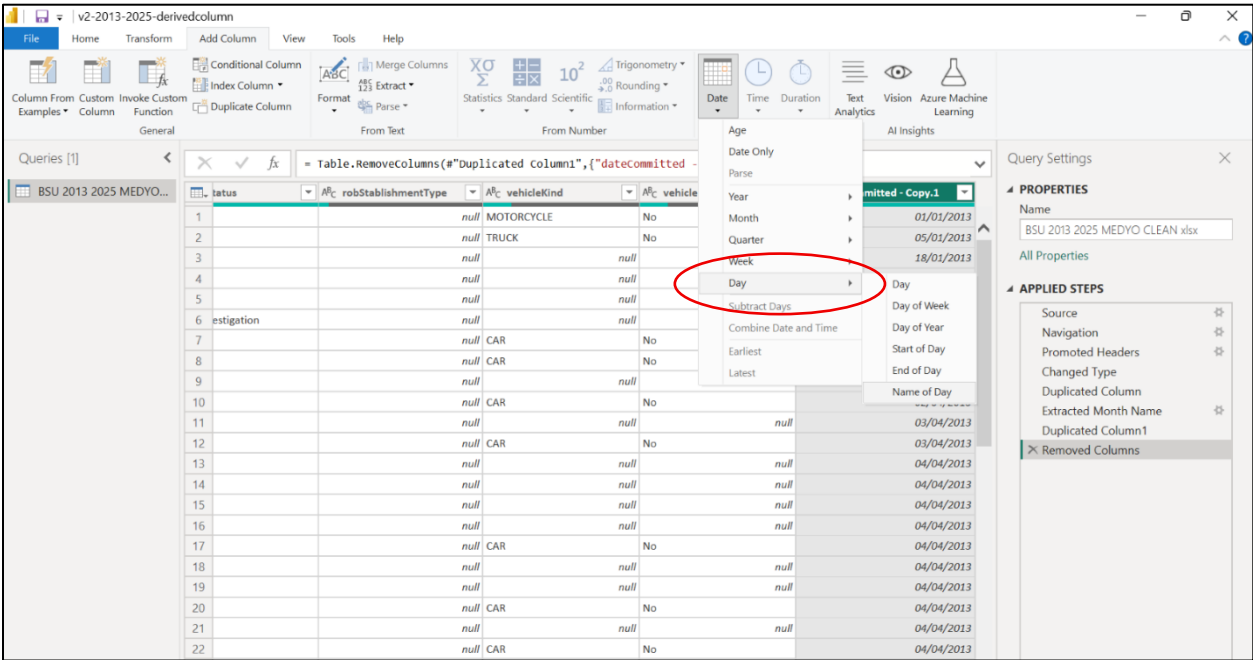


Using the "Fill down" method to copy values from the cell above into empty cells with the same Blotter No. The value in the first cell is automatically copied down to the cells below, making all rows under that column share the same Blotter No. without needing to manually enter it. This helps keep the data complete after editing or splitting rows.

POWER BI

Figure 8

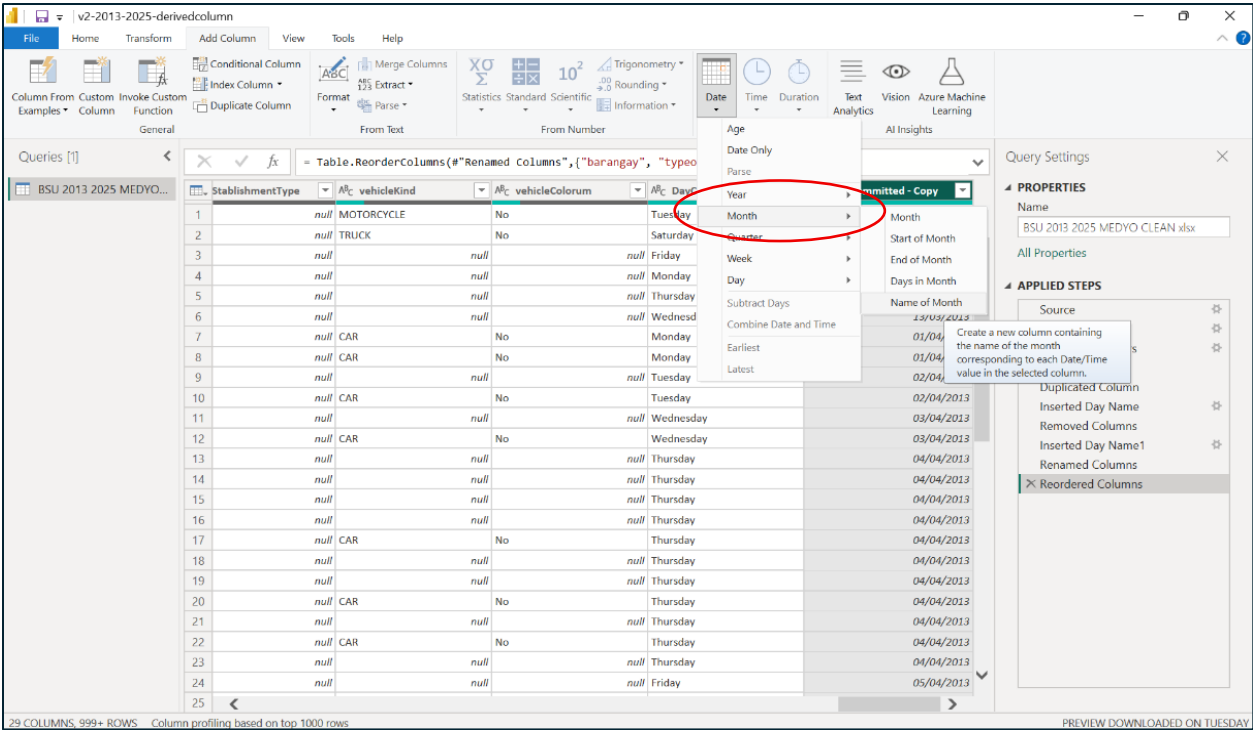
Derived Attribute (Day Committed)



The figure shows the derived attribute of *DayCommitted*, created from the attribute *DateCommitted*. This attribute will be helpful in analyzing trends and patterns in the daily occurrence of accidents in the Municipality of Calumpit.

Figure 9

Extracting MonthComitted from DateComitted

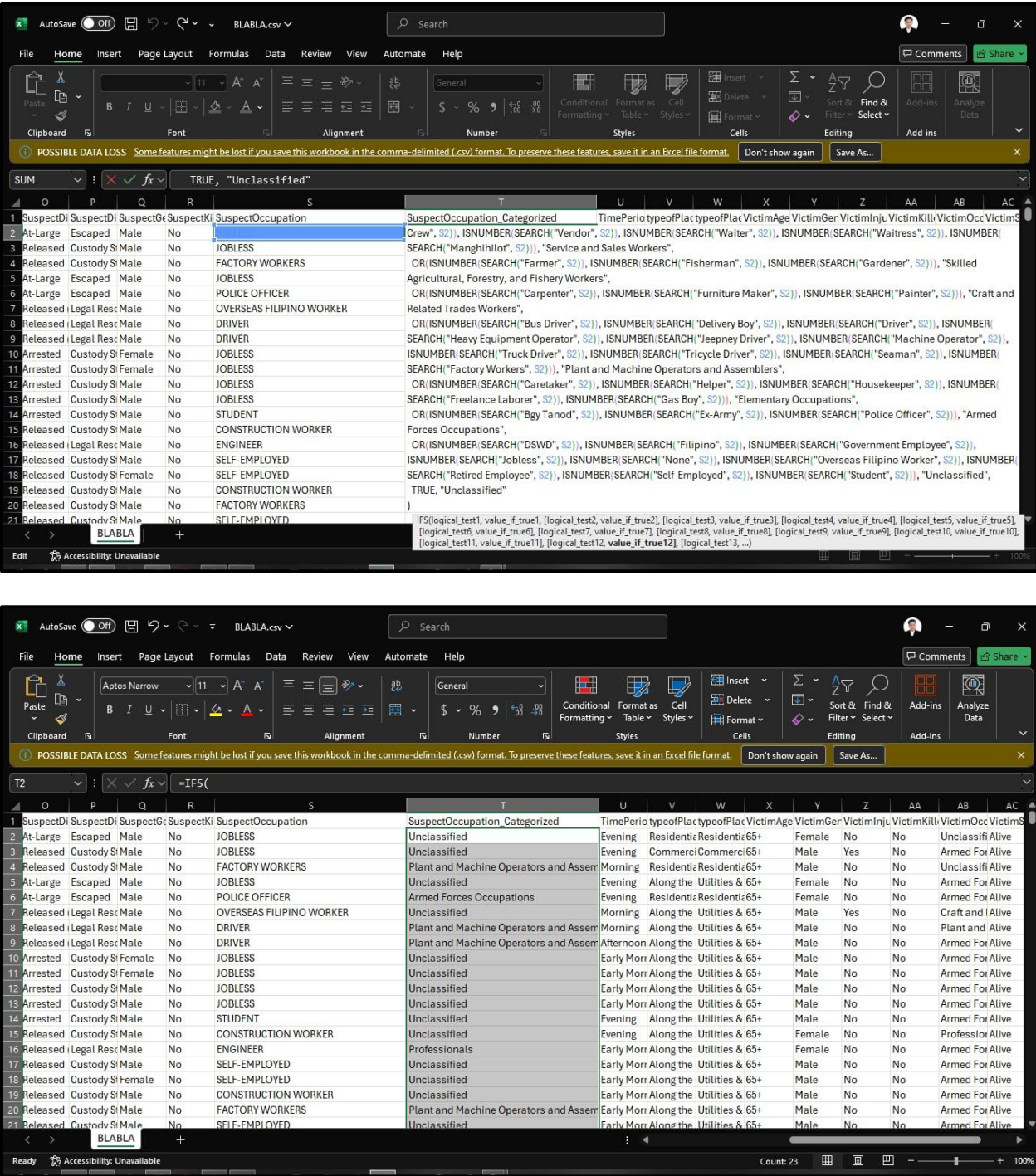


In Figure 3.1, the column *MonthCommitted* was extracted from the attribute *DateCommitted*. By utilizing this attribute, it will be possible to provide an insightful overview to determine which months have higher or lower rates of accidents in Calumpit.

Excel

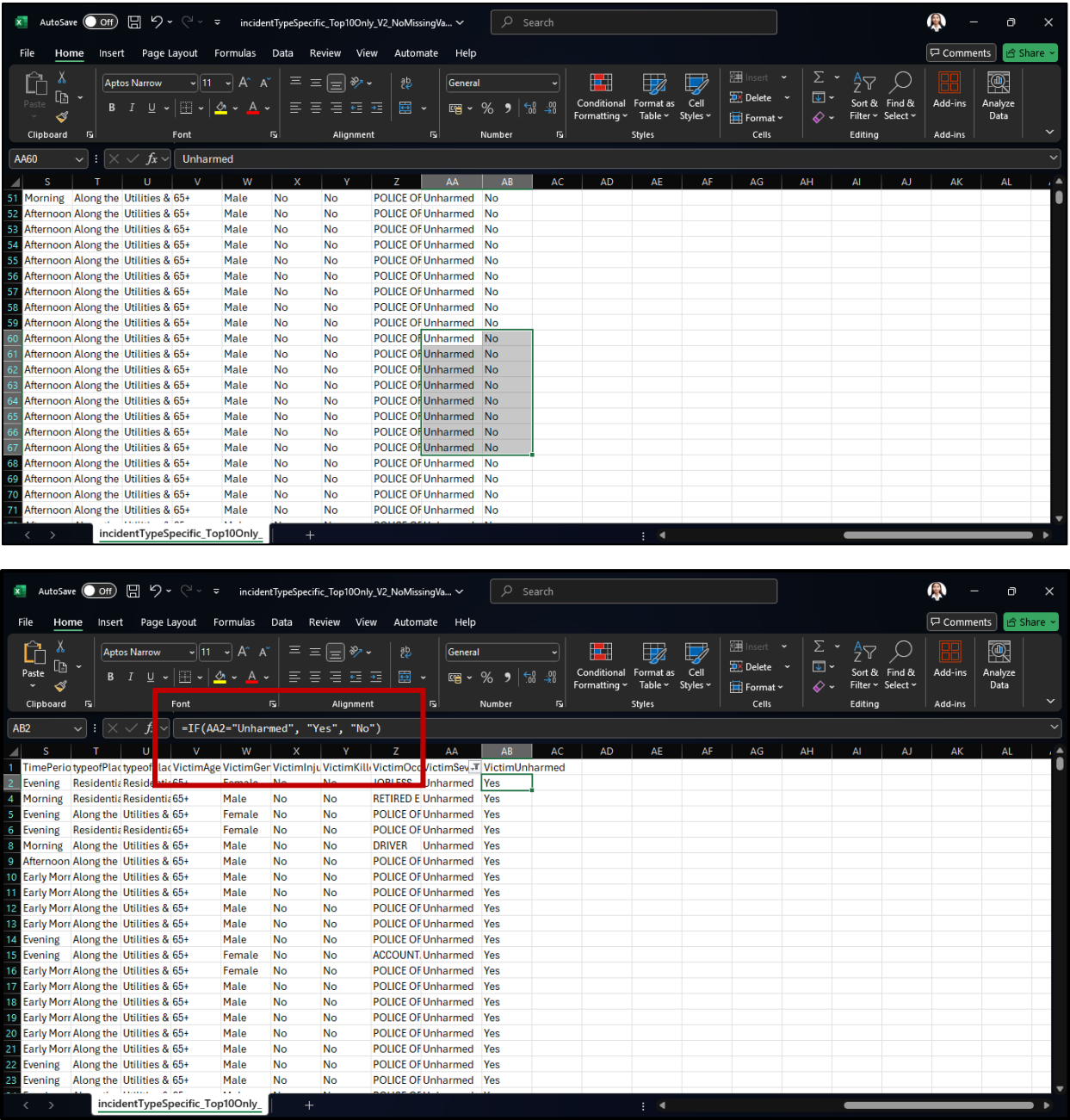
Figure 10

Binning or Categorizing values



In this figure, the Excel sheet shows a formula being used to classify suspect occupations into broader categories. The SuspectOccupation_Categorized column uses an IFS() function with SEARCH() and ISNUMBER() to match keywords in the SuspectOccupation field (e.g., “Driver”, “Jobless”, “Police Officer”) and assign them to generalized groups like “Unclassified”, “Armed Forces Occupations”, or “Plant and Machine Operators”. In the same way, the following variables are also categorized for uniformity and to control unique values: IncidentTypeSpecific, typeofPlace, robStablishmentType, and SuspectDisposition.

Figure 12
Filtering irrelevant or noisy records & Standardizing VictimUnharmed



In this figure, the data cleaning process standardizes the `VictimUnharmed` field by converting every record where `VictimSeverity` equals “Unharmed” into “Yes” and all other records into “No.” This transformation aligns the two related fields, removes ambiguity, and ensures that the dataset uses a clear, binary format for analysis. By applying this rule across all columns, inconsistencies are eliminated, and the data becomes more reliable for downstream reporting or modeling.

Figure 13
Standardizing SuspectAtLarge

incidentTypeSpecific_Top10Only																			
Search																			
File Home Insert Page Layout Formulas Data Review View Automate Help																			
Clipboard Font Alignment Number Styles Cells Editing Add-ins																			
M3 fx No																			
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	barangay	casestatus	DayComm	IncidentType	IncidentType	IncidentType	modus	MonthCon	offenseType	robStabilis	stageoffel	SuspectArrested	SuspectAtLarge	SuspectDi	SuspectGe	SuspectKi	SuspectOv	TimePerio	typeof
2	BALITE	Cleared	Monday	Incident	Violence	Domestic	Psycholog	June	Republic / ?	CONSUMM	25-64	No	Yes	At-Large	Male	No	JOBLESS	Evening	Reside
3	BALITE	Cleared	Sunday	Incident	Physical	Assault / ?	Punching	September	Crimes Ag	?	CONSUMM	25-64	No	No	Released	Male	No	?	Evening
4	BALITE	Cleared	Sunday	Incident	Robbery	Highway	Pushing	June	Crimes Ag	Resident	CONSUMM	25-64	No	No	Released	Male	No	?	FACTORY
5	BALITE	Cleared	Sunday	Others	Vehicular	Accident	(?	March	Republic / ?	CONSUMM	25-64	No	Yes	At-Large	Male	No	?	Evening	Along
6	BALITE	Cleared	Tuesday	Incident	Violence	Domestic	Psycholog	July	Republic / ?	CONSUMM	25-64	No	Yes	At-Large	Male	No	POLICE	OF	Evening
7	BALITE	Solved	Friday	Incident	Physical	Assault / ?	Mauling	December	Crimes Ag	?	CONSUMM	25-64	No	No	Released	Male	No	OVERSEAS	Morning
8	BALITE	Solved	Friday	Incident	Vehicular	Accident	(?	December	Crimes Ag	?	CONSUMM	25-64	No	No	Released	Male	No	DRIVER	Morning
9	BALITE	Solved	Friday	Operation	Buy Bust	Buy-Bust	(Pushing	October	Republic / ?	CONSUMM	25-64	No	No	Released	Male	No	DRIVER	Afternoon	Along
10	BALITE	Solved	Friday	Operation	Illegal Ga	Illegal Ga	Pushing	July	President	?	CONSUMM	25-64	Yes	No	Arrested	Female	No	?	Early Morri
11	BALITE	Solved	Friday	Operation	Illegal Ga	Illegal Ga	Pushing	July	President	?	CONSUMM	65+	Yes	No	Arrested	Female	No	?	Early Morri
12	BALITE	Solved	Friday	Operation	Illegal Ga	Illegal Ga	Pushing	July	President	?	CONSUMM	25-64	Yes	No	Arrested	Male	No	?	Early Morri
13	BALITE	Solved	Friday	Operation	Illegal Ga	Illegal Ga	Pushing	July	President	?	CONSUMM	25-64	Yes	No	Arrested	Male	No	?	Early Morri
14	BALITE	Solved	Friday	Others	Vehicular	Accident	(?	February	Crimes Ag	?	ATTEMPT	25-64	Yes	No	Arrested	Male	No	STUDENT	Evening
15	BALITE	Solved	Saturday	Incident	Vehicular	Accident	(?	July	Crimes Ag	?	CONSUMM	25-64	No	No	Released	Male	No	CONSTRUI	Evening
16	BALITE	Solved	Saturday	Incident	Vehicular	Accident	(?	October	Crimes Ag	?	CONSUMM	25-64	No	No	Released	Male	No	ENGINEER	Early Morri
17	BALITE	Solved	Saturday	Operation	Buy Bust	Buy-Bust	(Possessic	March	Republic / ?	CONSUMM	25-64	No	No	Released	Male	No	SELF-EMPI	Early Morri	Along
18	BALITE	Solved	Saturday	Operation	Buy Bust	Buy-Bust	(Possessic	March	Republic / ?	CONSUMM	25-64	No	No	Released	Female	No	SELF-EMPI	Early Morri	Along
19	BALITE	Solved	Saturday	Operation	Buy Bust	Buy-Bust	(Possessic	March	Republic / ?	CONSUMM	25-64	No	No	Released	Male	No	CONSTRUI	Early Morri	Along
20	BALITE	Solved	Saturday	Operation	Buy Bust	Buy-Bust	(Possessic	March	Republic / ?	CONSUMM	25-64	No	No	Released	Male	No	FACTORY	Early Morri	Along
21	BALITE	Solved	Saturday	Operation	Buy Bust	Buy-Bust	(Possessic	March	Republic / ?	CONSUMM	25-64	No	No	Released	Male	No	SELF-EMPI	Early Morri	Along

AutoSave

incidentTypeSpecific_Top10Only_V2_NoMissingVal...

Search

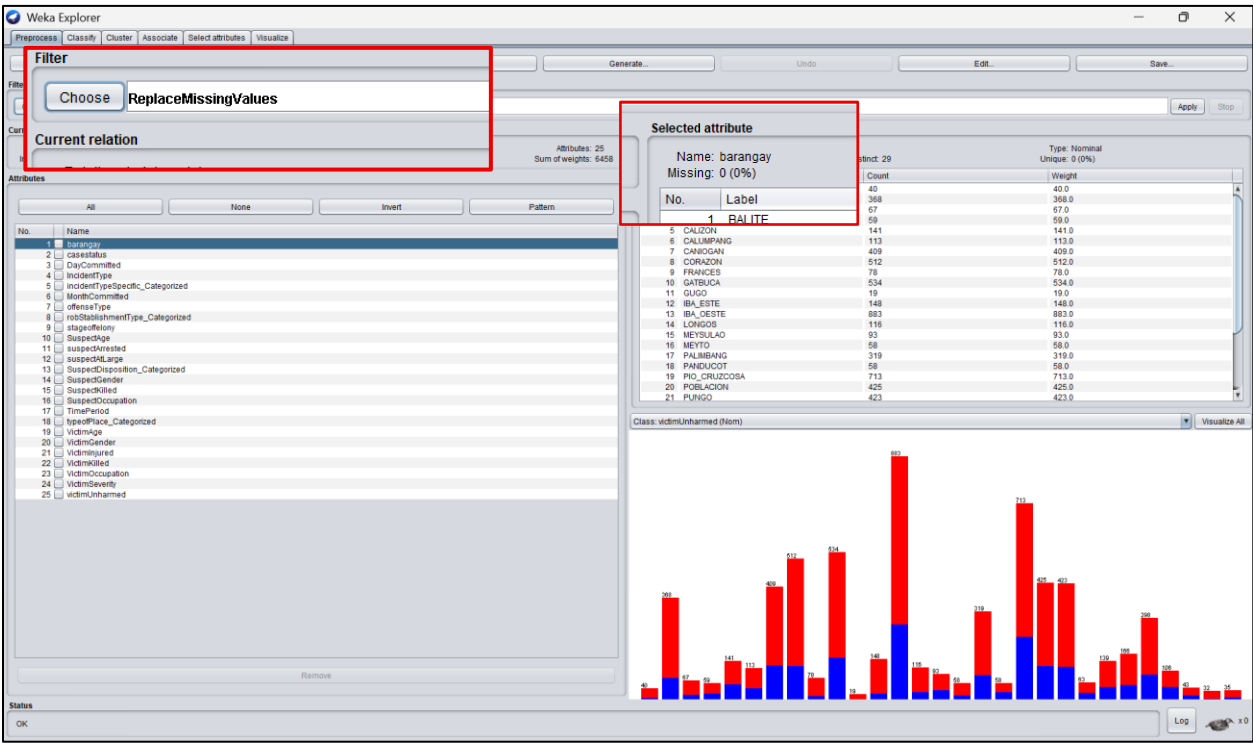
FileHomeInsertPage LayoutFormulasDataReviewViewAutomateHelp

ClipboardFontAlignmentNumberConditional FormattingFormat as TableCell StylesInsertDeleteFormatCellsEditingAdd-insAnalyze Data

N2

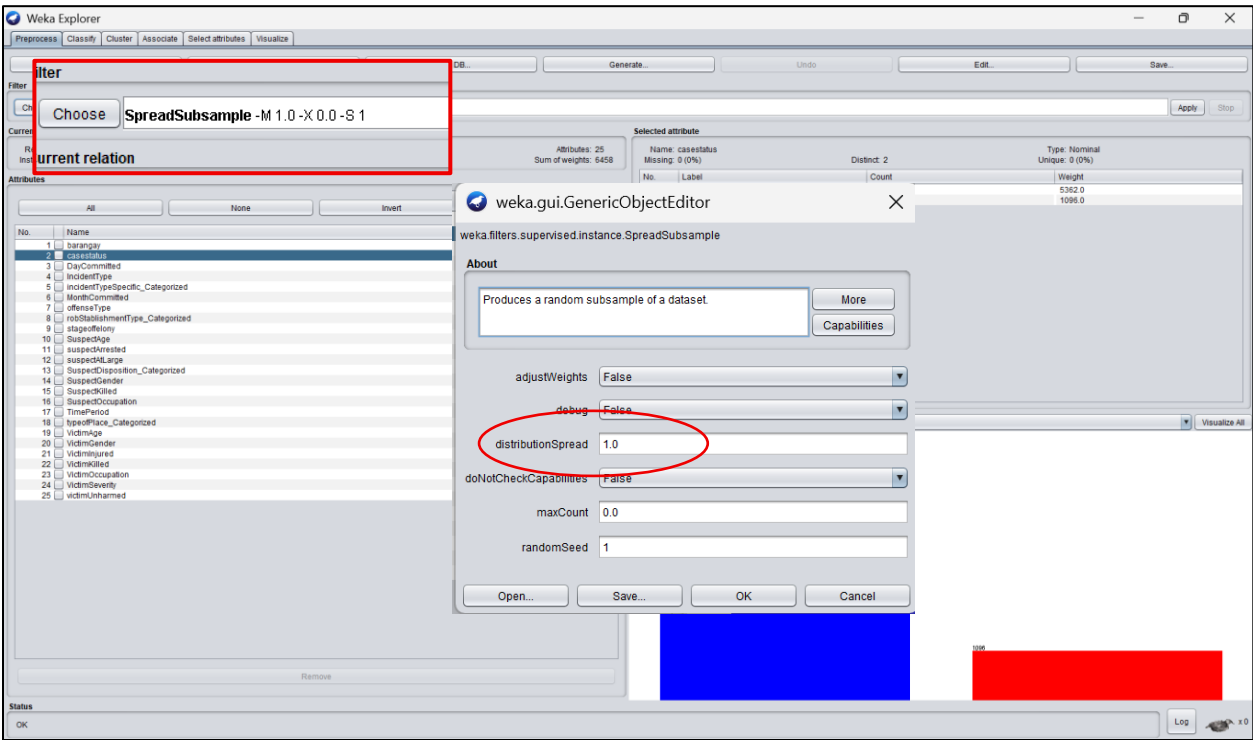
In the provided figure, the data cleaning process provides consistency between the columns SuspectArrested and SuspectAtLarge. The formula applied checks whether SuspectArrested equals "No," and if so, it puts "Yes" into SuspectAtLarge, signifying that the suspect is at large. If SuspectArrested is not "No" (that is, the suspect has been arrested), it puts "No" into SuspectAtLarge. This correction makes the connection between the two columns official, thus ensuring that the status of a suspect being at large is correctly represented based on their arrest status. The outcome gives cleaner and more dependable data for additional analysis

Figure 14
Imputation of Missing Values



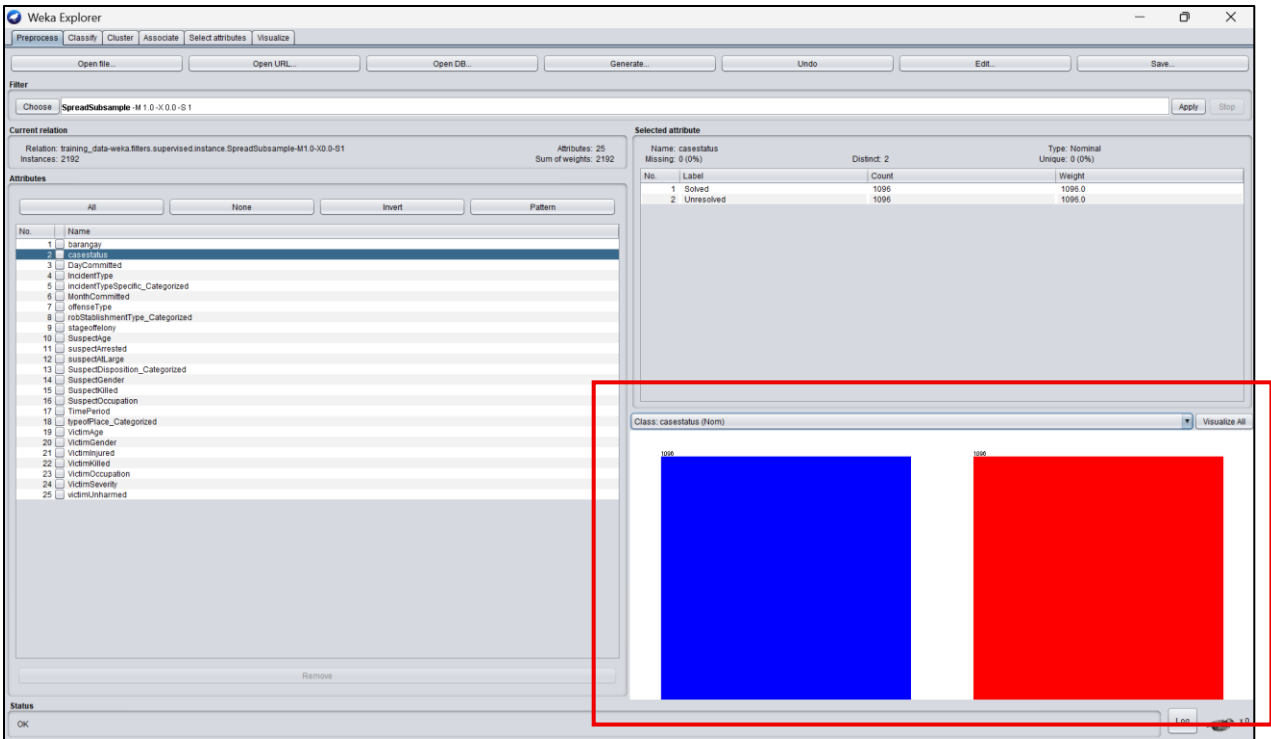
The figure shows the imputation of values using WEKA's ReplaceMissingValues filter in the preprocess tab. The filter replaces the dataset's (all columns but skips the target class) nominal and numerical values with the modes and means of the columns.

Figure 15
SpreadSubSample the Target Variable



The figure shows that we used RandomSpreadSubsample and set the distribution spread to 1.0 to balance the dataset. This parameter controls the class distribution after resampling. By setting it to 1.0, we ensured an even distribution among classes.

Figure 16
Balanced Target Class

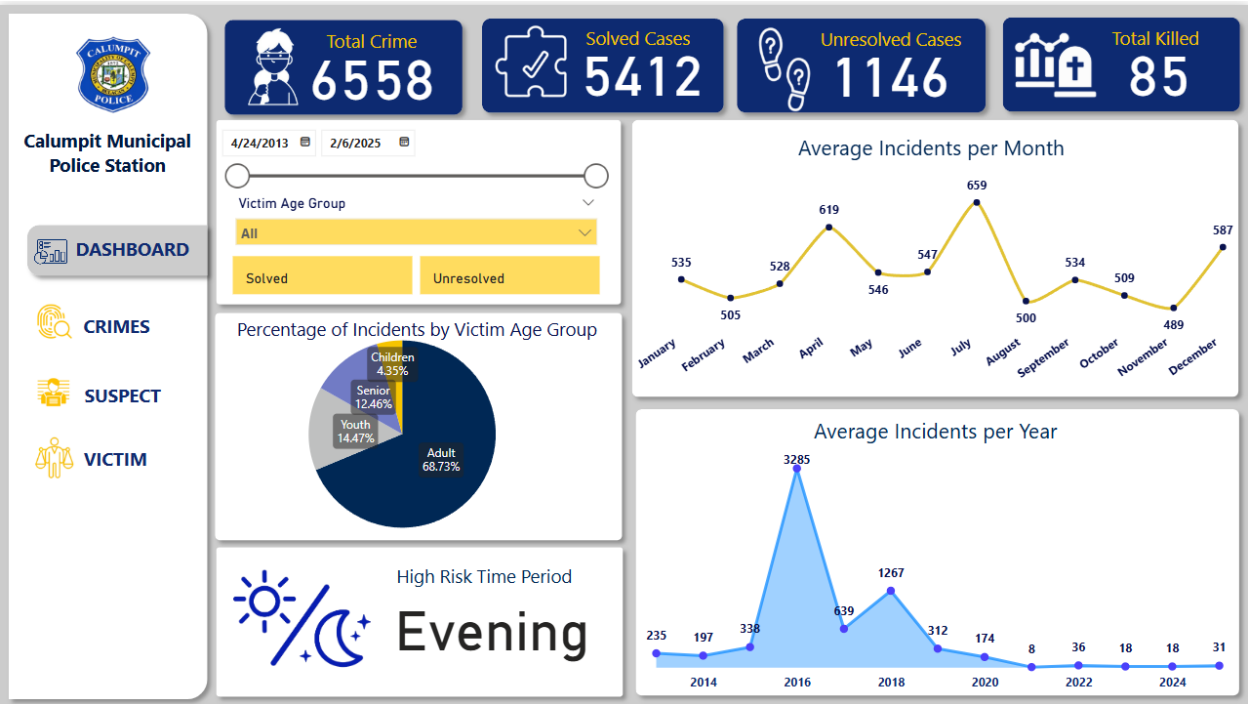


This figure shows that after applying RandomSpreadSubsample, the class distribution for 'case status' was balanced, with each class having 1,096 instances.

Dashboard & Insights

Figure 17

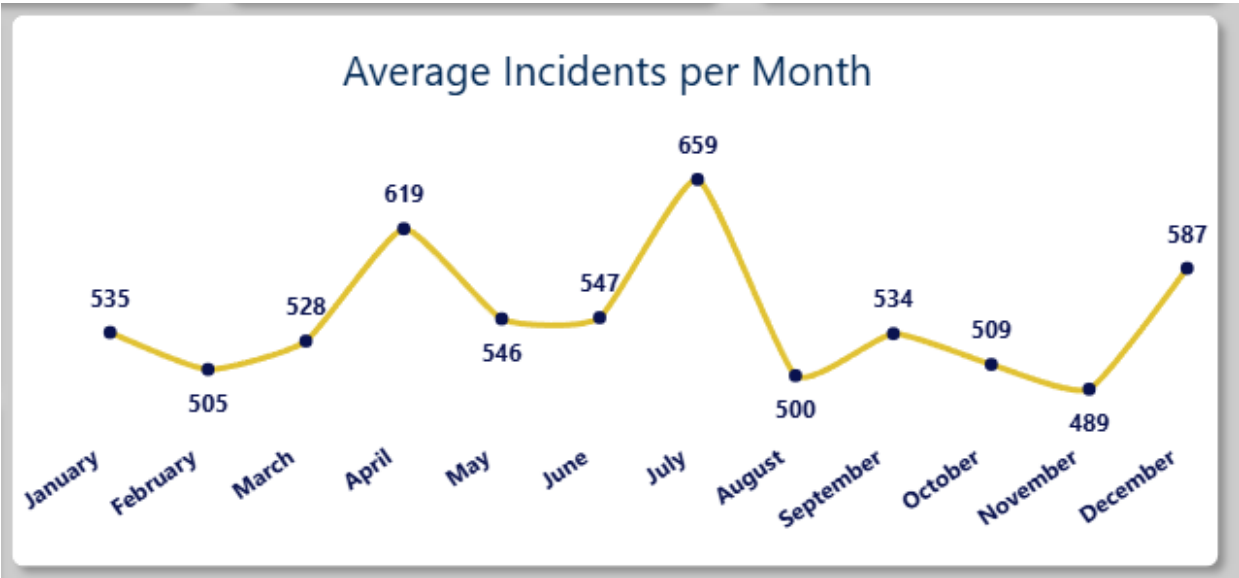
CMPS Dashboard Tab



The Dashboard Page of Calumpit Municipal Police Station presents the professional overview and comprehensive crime data from April 2013 to February 2025. The Card that shows the total crime, total number of solved cases, total number of unresolved cases, and total killed provides a clear value and the needs for continued focus on both investigative work and preventive measures. The slicer part indicates for the case status, victim age group, and the crime date committed, this allows the stakeholder to refine data based on specific timeframe and victim age demographics. The reported crimes achieve a total of 6,558, with the highest number of solved cases, about 5,412, which reflects an effective policing effort, through the presence of 1,146 unresolved cases and 85 total fatalities highlighting areas that require continuing attention. The percentage of Incidents by Victim Age Group shows that Adults were majority of the victims (68.73%), highlighting further targeted prevention strategies, while the incident peaks showed during the month of July and evening hours, which indicates the need for more deployment of police officer. Additionally, the average incident per year surge in 2016, emphasize the highest number of incidents happen during the mentioned year.

Figure 18

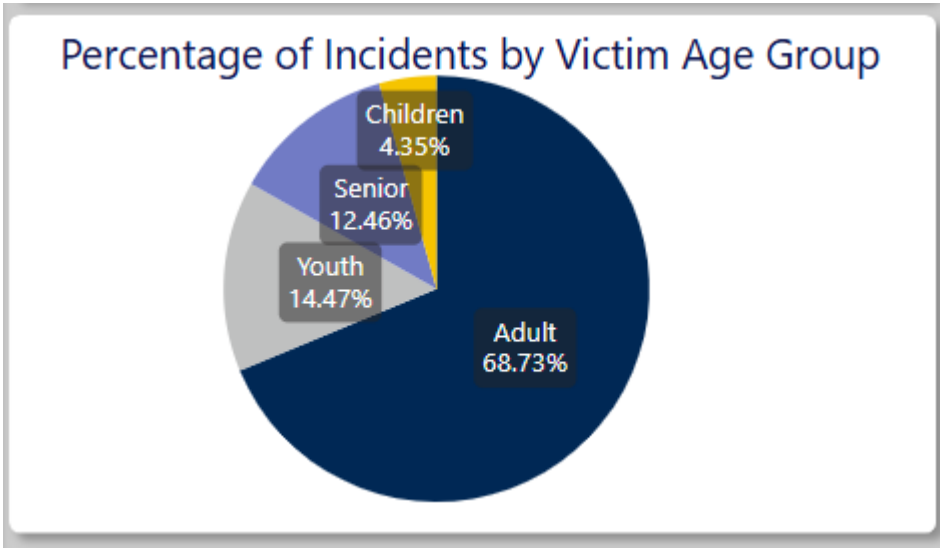
CMPS Dashboard - Average Incident Per Month



This line graph shows the calculated average number of incidents per month from the column incidentTypeSpecific_Categorized. With an average of 659 incidents, the month of July is revealed as the peak month which suggests possible event-related or seasonal spikes. It shows that in the month of November (489), it appears to be the lowest average of incidents from 2013 to 2025. In line with this, high authorities could guide resource allocation during high-risk month. Understanding these patterns is important for adjusting the patrol schedules and launching public safety campaigns to decrease the number of incidents during this peak month.

Figure 19

CMPS Dashboard - Percentage of Incident by Victim Age Groups



The visuals showing that adults were the common victims of an incident having a highest percentage among the age-group. It's most likely because adults are frequently presence in public spaces and active in workspace, which has a high possibility in risk environments. Even though

adults have the highest average, there are still vulnerabilities in youths, seniors, and children. Given these insights will help the law enforcement and local government to identify which age-group based required more focused on awareness and community engagement to decrease victimization and enhance public safety.

Figure 20

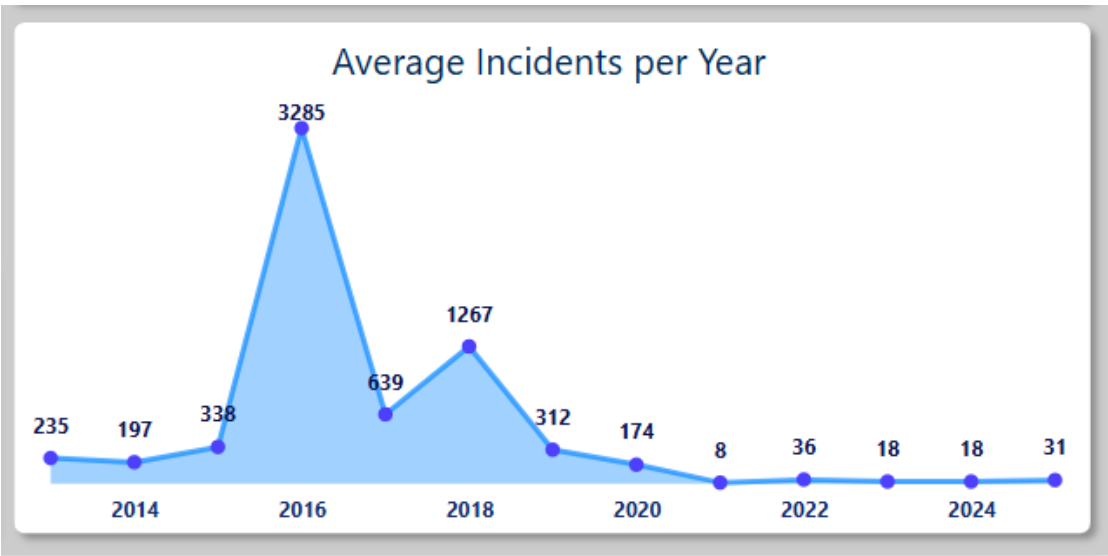
CMPS Dashboard - High Risk Time Period



In the figure it presents identifying what period the most frequent incidents happen. Evening, showed as the high-risk period which suggests incidents more likely to occur. These insights may be attributed to various factors, including alcohol-related offenses, social gatherings, and other factors that might happen during evening period. In line with this, local government and law enforcement could effectively deploy resources by increasing patrols and enhancing street lighting to ensure the safety of the community during peak hours.

Figure 21

CMPS Dashboard - Average Incident Per Year

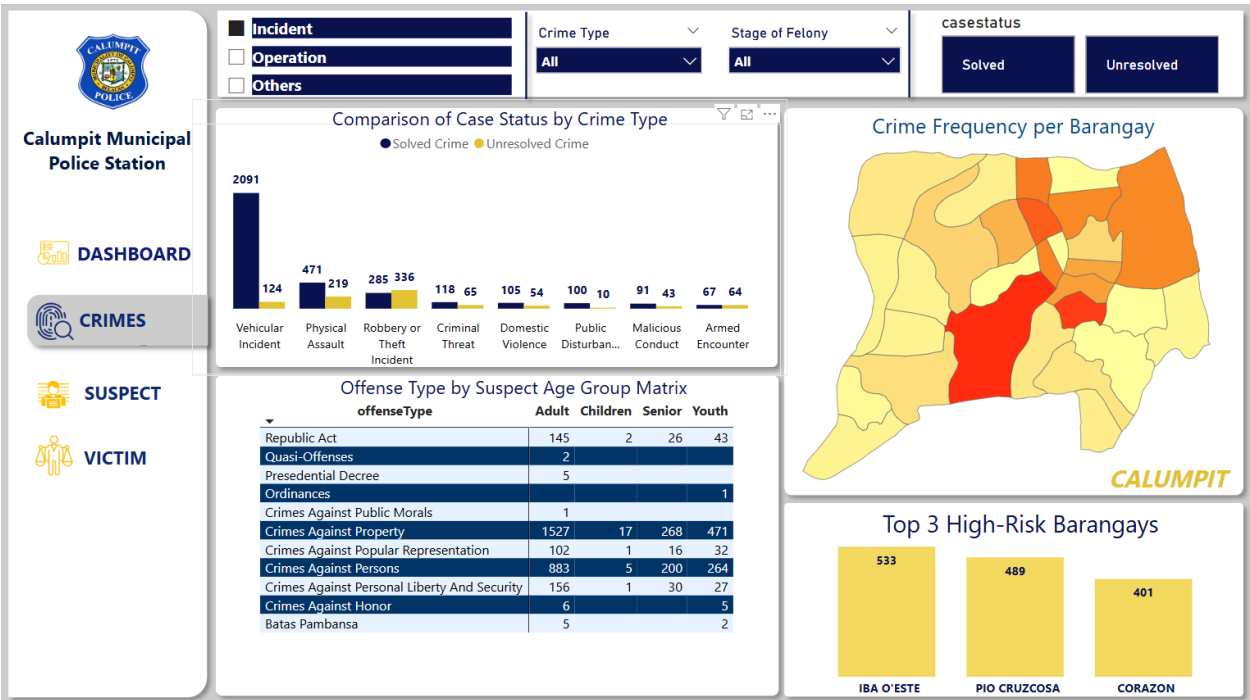


This Line Graph visual presents the highest average incidents per year from 2013-2025. The process used in Average Incident per Year is generated from the YearCommitted field and

counting incidents from incidentTypeSpecific_Categorized to get the total average. During these years, it is revealed that in the year 2016 an increased number of incidents happen in the Municipality of Calumpit. There are factors that may cause why this year marks as peak year in the recorded datasets such as economic hardships or gaps in law enforcement coverage. Given these insights could help the local government to identify long term trends, further investigate to allocate police officers or barangay patrols more effectively which may result in decreasing the number of incidents in the future.

Figure 22

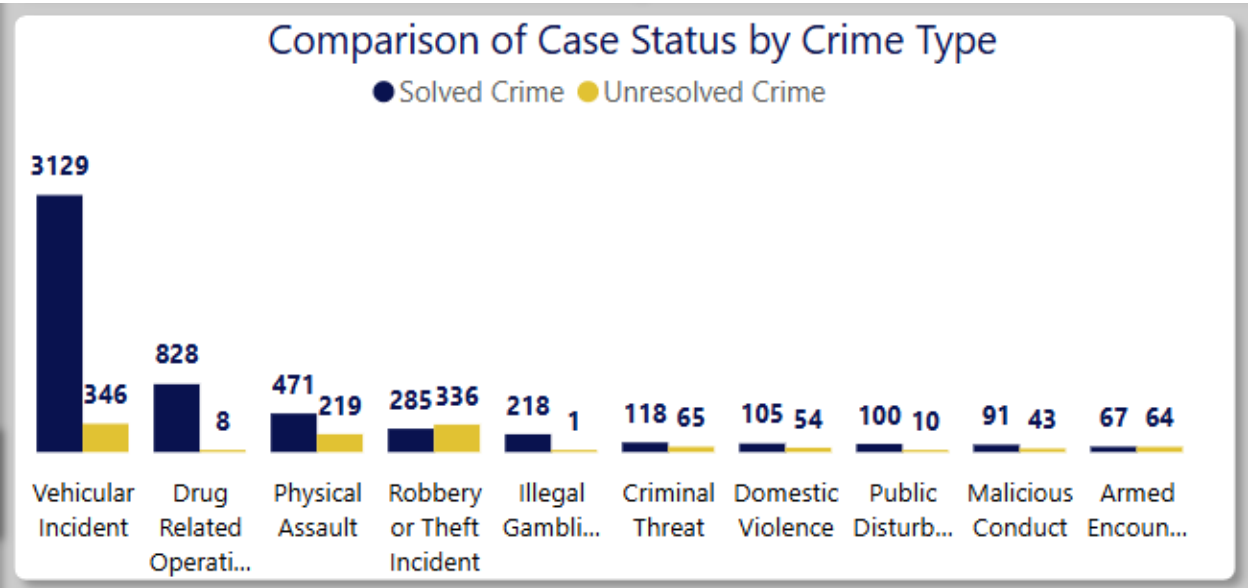
CMPS Crimes Tab



The crimes tab is a strategic component of the dashboard, offering an at-a-glance overview of criminal activity in Calumpit. The slicer for incident and crime type allows users to isolate and analyze patterns for specific crimes, identify which barangays are more affected by incident type, and support scenario-based planning. The button slicer for case status is crucial for police effectiveness and encourages transparency and progress tracking in police case management. Lastly, the stage of felony adds legal depth to the crime assessment as it supports legal coordination and understanding of the nature of crimes per area of demographic. All in all, the visual clarity, interactivity, and drill-down capabilities empower law enforcement officers and investigators to monitor trends, measure case resolutions, and focus on crime hotspots.

Figure 23

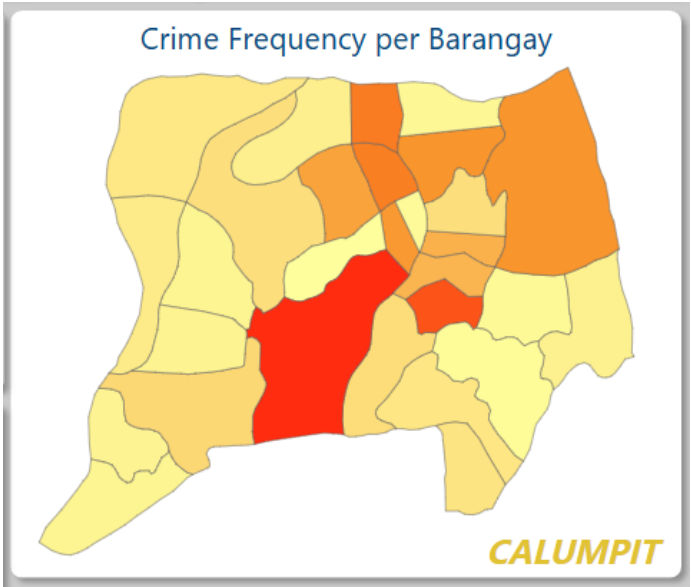
CMPS Crimes - Comparison of Case Status by Crime Type



Presenting the comparison of case status by crime type in Calumpit using bar chart, distinguishing between solved and unresolved cases. From the figure above, it reveals that Vehicular Incident have the highest solved cases with 3,129 and 346 unresolved cases indicating that it is the most common incident occurred in Calumpit. On the other hand, Drug-Related operation followed next with 828 solved cases, but surprisingly, only 8 remain for unresolved cases, this indicates an effective handling such cases. With these insights, it highlights which case needs further investigation for unresolved cases and maintain the solved cases.

Figure 24

CMPS Crimes - Crime Frequency per Barangay



This chart illustrates a shape map of Calumpit, Philippines, displaying crime frequency by barangay, labels for each barangay will appear on hover because displaying them permanently

would overcrowd the map and reduce readability. The map has a color scale from yellow (low crime frequency) to dark red (high crime frequency). The darker the red, the higher the crime rate barangays are. This space visualization of crime data is vital in locating the high-crime areas, maximizing resource allocation, and guiding focused crime prevention efforts.

Figure 25

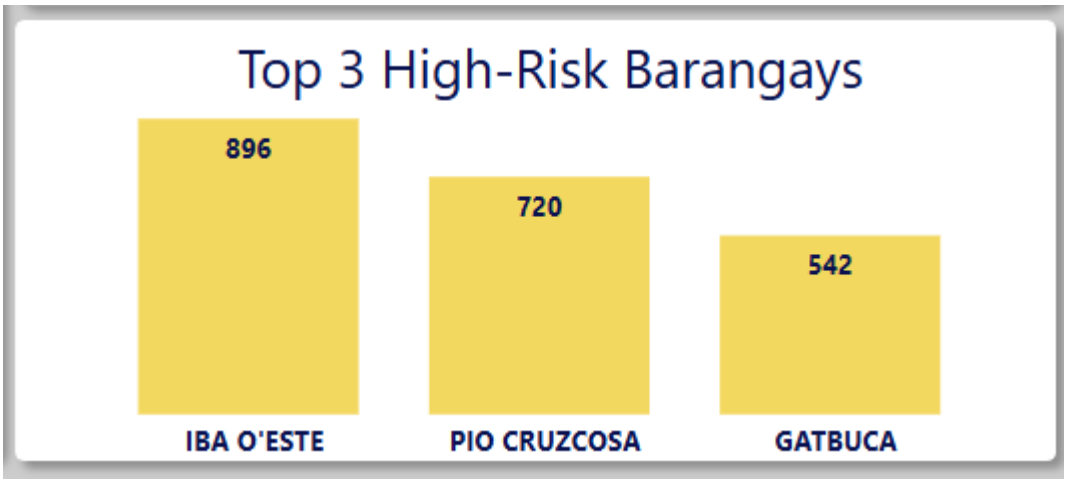
CMPS Crimes - Offense Type by Suspect Age Group Matrix

Offense Type by Suspect Age Group Matrix				
offenseType	Adult	Children	Senior	Youth
Republic Act	882	8	181	270
Quasi-Offenses	2			
Presidential Decree	19	1	1	
Ordinances	3			3
Crimes Against Public Morals	1			
Crimes Against Public Interest	5			
Crimes Against Property	2140	20	375	584
Crimes Against Popular Representation	116	1	17	40
Crimes Against Persons	1079	8	226	303
Crimes Against Personal Liberty And Security	171	1	37	30
Crimes Against Honor	8			5
Crimes Against Chastity	7		3	2
Batas Pambansa	7			2

This table displays a matrix indicating the suspect's frequency of types of offense by age category in the CMPS Crimes. The rows are various categories of offense (Republic Act, Quasi-Offenses, Presidential Decree, Ordinances, and miscellaneous crimes against public morals, public interest, property, popular representation, persons, personal liberty and security, honor, and chastity, and Batas Pambansa). Columns are offense categories: Drug Offense, Violent Offense, Property Offense, etc., and age groups: Adult, Children, Senior, and Youth. Each cell contains the number of offenses in a specific offense category and age group. This analysis allows trends in criminal activity by offense type and age offending to be determined. The data allow more advanced understanding of crime patterns, which can result in more effective and targeted responses.

Figure 26

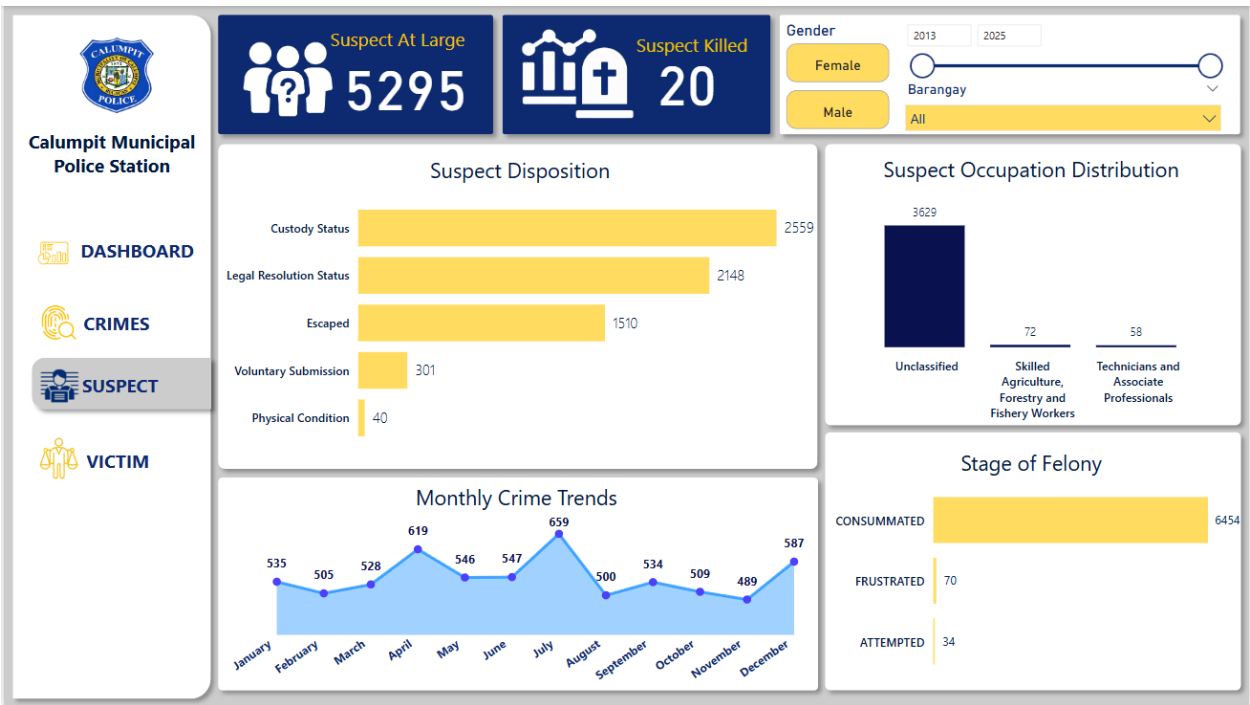
CMPS Crimes - Top 3 High-Risk Barangays



This bar chart presents the top 3 high-risk barangays in Calumpit based on the number of incidents recorded. The barangay of Iba O’Este showed with the highest rank for 2013-2025 record with 896 incidents, followed by Pio Cruzcosa with 720, and Gatbuca with 542. In line with this, Iba O’Este needs the most immediate attention and barangay patrols for risk mitigation among other barangays in Calumpit. Given this insightful visual will help the local government and barangay officials to develop an effective strategic plan for allocating patrols and prioritize high-risk with maintaining proactive measures in other barangay.

Figure 27

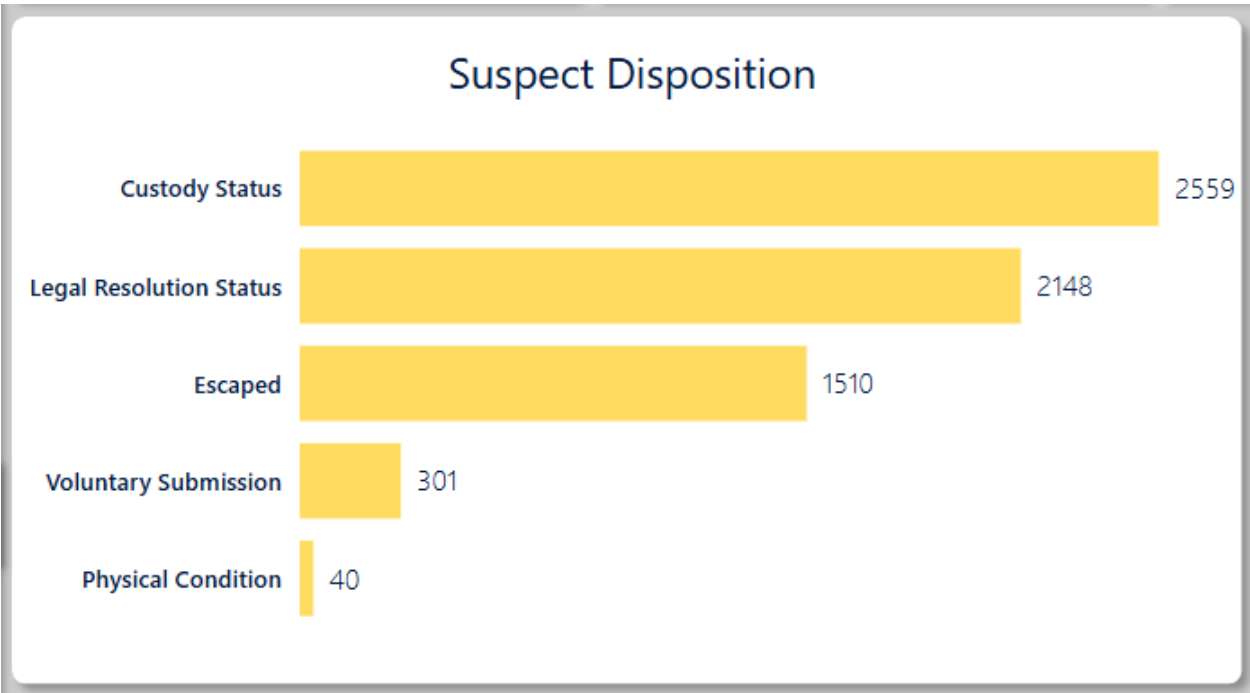
CMPS Suspect Tab



The CMPS dashboard's Suspect tab profiles Calumpit suspects through interactive filters (gender, barangay, date range) for focused analysis. Important indicators reveal suspects at large

and killed, while charts present suspect disposition, occupation (largely unclassified), and the stage of felony (largely consummated). A monthly crime trend graph displays fluctuations, which help detect patterns and possible enforcement gaps. This tab facilitates data-driven crime prevention and resource planning.

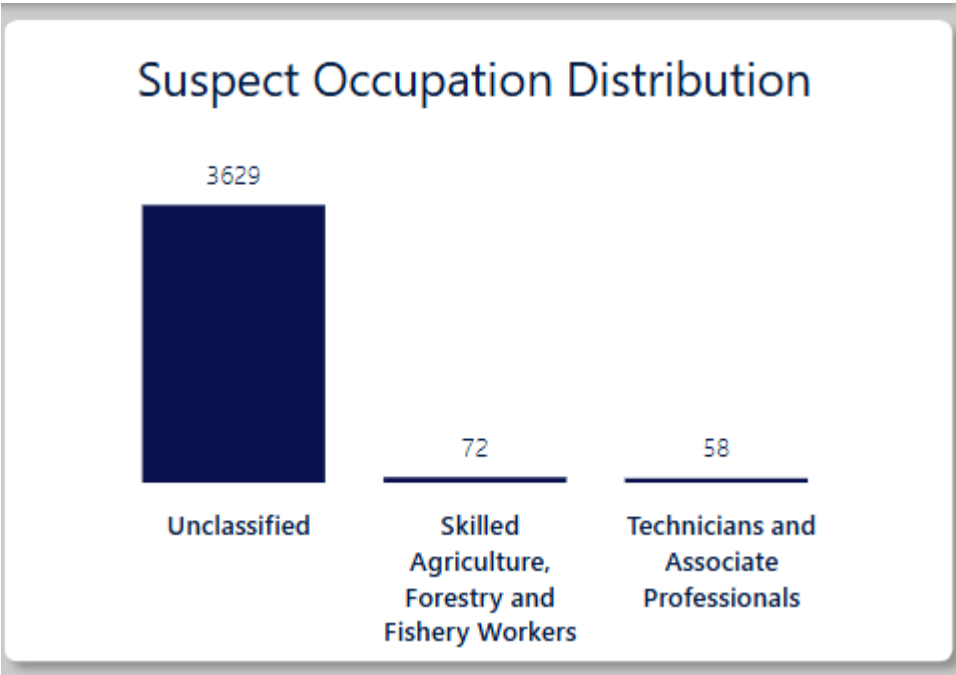
Figure 28
CMPS Suspect- Suspect Disposition



This bar chart illustrates the disposition of criminal incident suspects such as Custody Status (2559), Legal Resolution Status (2148), Escaped (1510), Voluntary Submission (301), and Physical Condition (40). The graph comfortably illustrates that the majority of suspects are in custody or have been brought to legal resolution, and a great percentage escaped. The comparatively small numbers for voluntary surrender and physical status suggest these are less frequent events. The graph here provides a quick snapshot of suspect statuses conducive to comparing law enforcement effectiveness and planning resources.

Figure 29

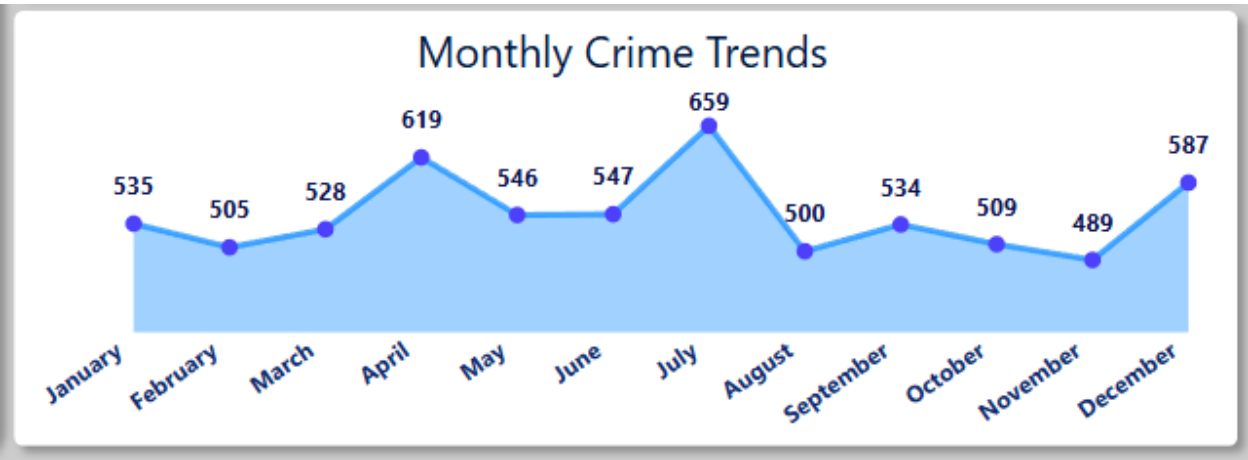
CMPS Suspect - Suspect Occupation Distribution



This bar graph shows the occupation distribution of suspects and indicates a broad spread. There is a vast majority (3629) that is "Unclassified" and relatively smaller quantities are accounted for in individual occupational categories such "Skilled Agriculture, Forestry and Fishery Workers," and "Technicians and Associate Professionals." This data reveals there is a need for improved collections of data for suspect occupations to enable more useful crime analysis and perhaps reveal occupational patterns of criminality.

Figure 30

CMPS Suspect - Monthly Crime Trends

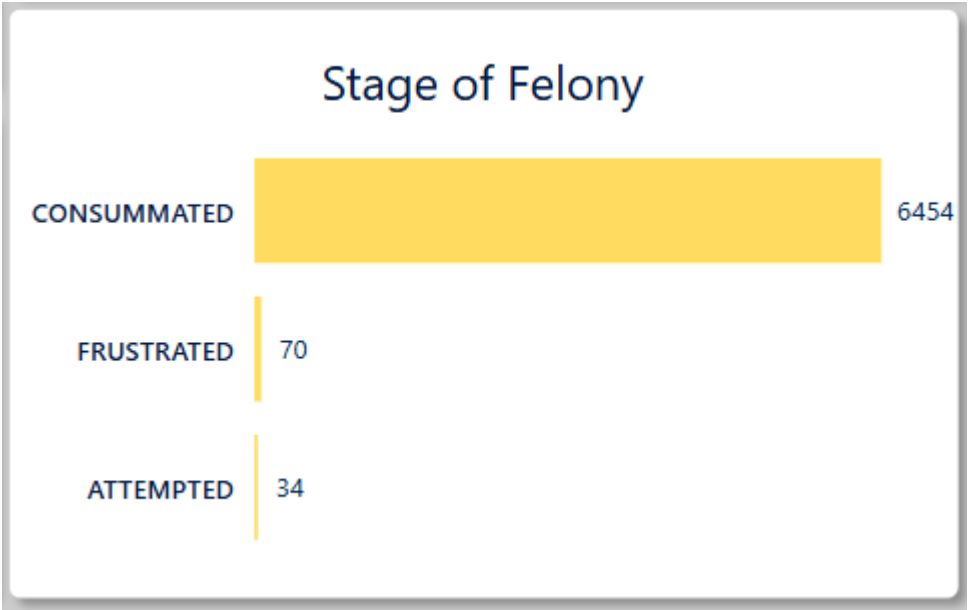


This monthly crime line graph is for crime trend analysis. The highest incidents of crime were in April and July, with 24 incidents for each of these months. In November, there was a dip to 18 incidents. This graph presents a time-series view of various crime activity characters, which

would be useful to potential discovery of seasonal patterns or patterns of times of the year that must be targeted by law enforcement.

Figure 31

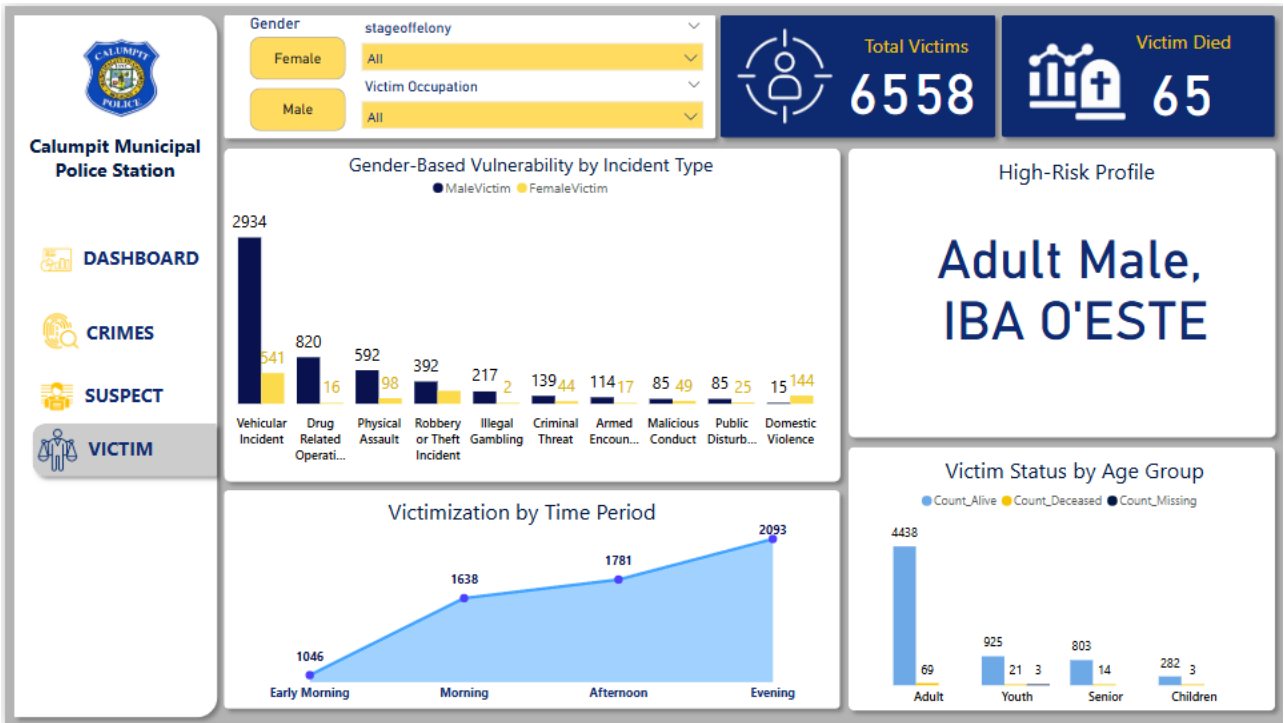
CMPS Suspect - Stage of Felony



This bar chart presents data on the stage of felony for criminal cases within the CMPS. This bar chart shows a large disparity in felony stages with most crimes (6454) consummated, far exceeding frustrated (70) and attempted (34) cases. This implies that the majority of crimes advance to their ultimate stages, showing that the crime-control measures need to be more proactive and in the early stages. The facts point to possible areas of compromise in crime deterrence and swift response.

Figure 32

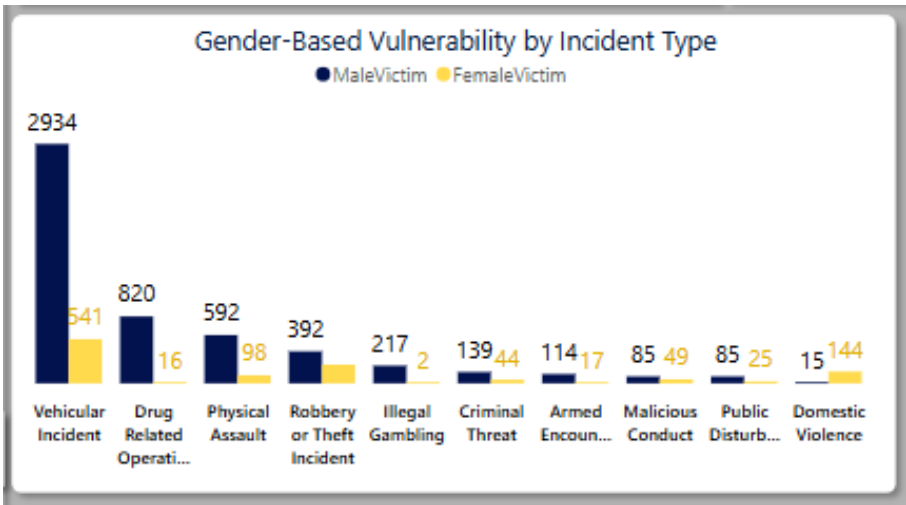
CMPS Victim Tab



The victim tab of the CMPS dashboard allows the users to shift their focus primarily on the information of victims that were involved in incidents. It features 3 slicers, for gender, stage of felony, and incident type. This interactivity enhances the analytical depth allowing for gender-specific vulnerability analysis, crime trend analysis by felony stage, and more customized filtering by incident type. Moreover, the two summary cards at the upper right of the figure offer at-a-glance key performance indicators (KPI) to provide context about the scale of victimization across Calumpit. These indicators allow the police to analyze number of victims against the fatality rate of each incidents. Particular use case scenario for this is if a user wants to focus on male victims involved in crimes at the “Attempted” stage to identify patterns and emerging threats. With its strong focus on victim-centered analytics, this tab supports the broader goal of improving community safety and victim support initiatives.

Figure 33

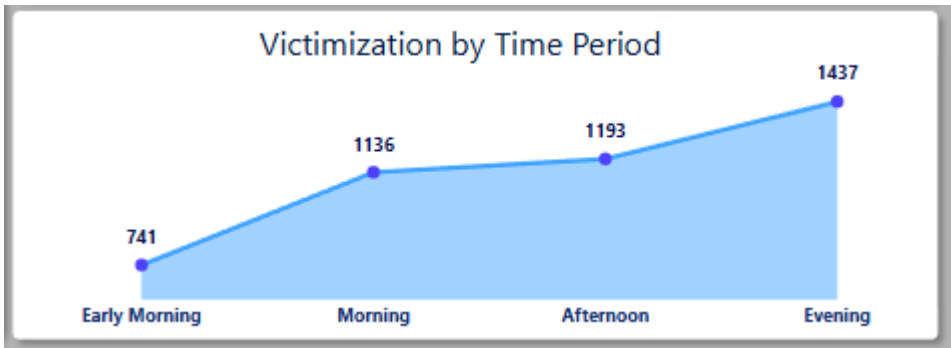
CMPS Victim Tab – Gender-Based Vulnerability by Incident Type



This clustered column chart compares the number of male vs female victims across all the values of incident type. This visualization helps in identifying which demographics are most vulnerable to certain crimes. In the figure above, we can see that in almost all, except for one, incident types present, male victims significantly outnumber female victims. This suggests that male victims may have higher exposure to public or street-related threats. The prominence of female victims in domestic violence cases can guide community outreach, legal assistance, or protective services targeting domestic abuse.

Figure 34

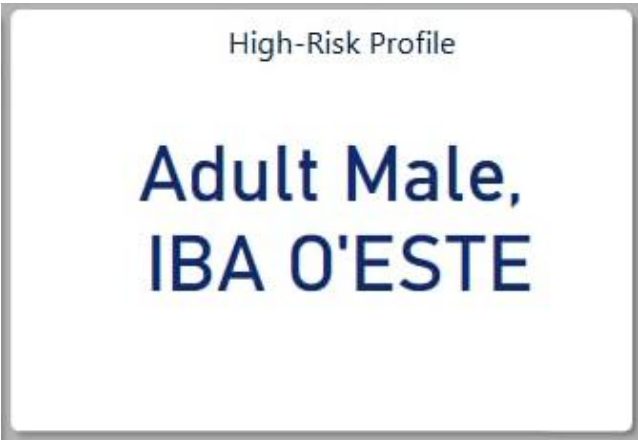
CMPS Victim Tab – Victimization by Time Period



This line graph shows the amount of victim cases or victimizations at various periods of the day. As we can see from the figure, it is an uptrend where early morning has the fewest (741) recorded victim cases with a sharp gradual increase up to evening having the highest (1437) percentage of recorded victim cases across all time periods. Respectively, the morning (1136) and afternoon (1193) fall in the center. The timing of police patrols and crime prevention tactics depend heavily on this information. The station can better manage resources and possibly lower victimization rates by detecting high-risk times.

Figure 35

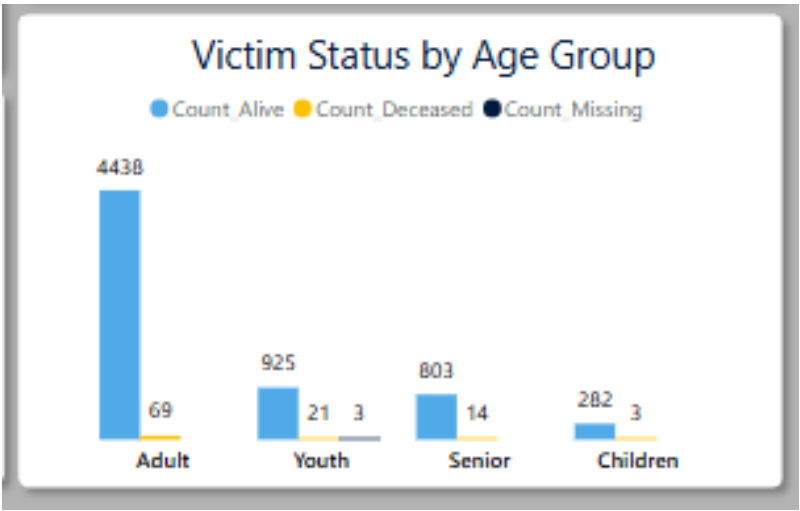
CMPS Victim Tab – High-Risk Profile



This card synthesizes the most common victim profile based on the dataset and the slicers applied. The figure shows the Victim Age Group (Adult), Victim Gender (Male), and the Barangay (Iba O’Este) showing the most vulnerable demographic and their common geographic location. Knowing this information would allow the police to tailor intervention programs, targeted awareness or safety campaigns by communicating and collaborating with barangay officials as well as strengthening visibility and presence in Iba O’Este.

Figure 36

CMPS Victim Tab – Victim Status by Age Group



The clustered column chart breaks down the status of victims across four key age groups, offering a more victim-centered analysis vital in understanding who the victims are, not just by count but by impact severity per age group. From the figure above, Adults (aged 25 - 64) form the largest group of victims with 4,438 survivors and 69 (1.53% fatality rate for adults) fatalities. This aligns with Figure 25, suggesting that adults have higher exposure to incidents due to mobility, employment and active presence in public spaces. This suggests that the Adults are not necessarily the most vulnerable, but are statistically at higher risk. Youth victims (aged 15-24) had 925 survivors, 21 fatalities (2.21% fatality rate for youths), and 3 missing cases. Being the second

highest age group for victimization shows strong vulnerability rate compared to Adults. Youth may be susceptible to drug-related operations, gang activity, or other incidents which can escalate into fatal scenarios. The presence of missing youth cases also introduces a critical concern about abduction, exploitation, or trafficking, instigating immediate investigation and intervention. Moving forward, senior citizens (aged 65 and over) form a modest portion of the dataset having 803 survivors and 14 fatalities (1.7% fatality rate in senior citizens). Seniors may be more vulnerable due to physical frailty, having slower response time or being targeted for theft or abuse. Lastly, while having the smallest group by count, Children victims (aged 0-14) had 282 survivors and 3 fatalities (1.05% fatality rate). Their inability to protect themselves poses a significant threat to their age groups. Intensive safeguarding efforts and the need for collaboration with child welfare agencies, schools and barangay level child protection units are needed.

Conclusion

Overall, the project successfully implements descriptive analytics by creating an interactive dashboard that will present an overview insight of incidents that happened from 2013-2025 in Calumpit, Bulacan. The Calumpit Municipal Police Station (CMPS) will benefit from the project, as well as the local government and barangay officials by understanding the most common incident that happens in the area. The Dashboard has 4-page features, including the dashboard page as the first page presenting the overview insights, followed by crimes page showing the drill down analysis of barangays incidents with heatmaps visuals for identifying which barangays has the frequent number of incidents then, the suspect and victim page highlighting the demographics insights. Ultimately, this visualization project will help the local government of Calumpit, Bulacan to effectively allocate law enforcement in high-risk areas to improve crime prevention and decrease the number of incidents in the future and for more secure and safety community.

References

- Authentication challenge pages. (2025). Psa.gov.ph. <https://psa.gov.ph/system/files?file=scd/files%20%281%29.pdf>
- CLUP HOUSING AND LAND USE REGULATORY BOARD. (n.d.). [https://dhsud.gov.ph/wp-content/uploads/Publication/Guidebooks/HLURB CLUP Vol 1.pdf](https://dhsud.gov.ph/wp-content/uploads/Publication/Guidebooks/HLURB_CLUP_Vol_1.pdf)
- FEMA US&R RESPONSE SYSTEM. (n.d.). Retrieved May 19, 2025, from <https://www.fema.gov/pdf/emergency/usr/unit9.pdf>
- Government of Canada, S. C. (1998, July 15). Age Categories, Life Cycle Groupings. Www.statcan.gc.ca. <https://www.statcan.gc.ca/en/concepts/definitions/age2>
- Khan, Muzammil, Ali, Azmat, Alharbi, Yasser, Predicting and Preventing Crime: A Crime Prediction Model Using San Francisco Crime Data by Classification Techniques, *Complexity*, 2022, 4830411, 13 pages, 2022. <https://doi.org/10.1155/2022/4830411>
- Saeed, Ruaa Mohammed and Abdulmohsin, Husam Ali. "A study on predicting crime rates through machine learning and data mining using text" *Journal of Intelligent Systems*, vol. 32, no. 1, 2023, pp. 20220223. <https://doi.org/10.1515/jisys-2022-0223>
- National Policy on the Management of the Dead and the 1 Missing Persons During Emergencies and Disasters 2 3 (*****This Manual covers only NATURAL disasters. (n.d.). Retrieved May 19, 2025, from [https://ndrrmc.gov.ph/attachments/article/1862/Management of the Dead and the Missing M DM with Line Numbers.pdf](https://ndrrmc.gov.ph/attachments/article/1862/Management%20of%20the%20Dead%20and%20the%20Missing%20M%20DM%20with%20Line%20Numbers.pdf)
- PHILIPPINE NATIONAL POLICE MANUAL REVISED PHILIPPINE NATIONAL POLICE OPERATIONAL PROCEDURES. (2021). <https://akg.pnp.gov.ph/wp-content/uploads/2024/01/POP-Manual-2021.pdf>
- Philippine Statistics Authority. (2023). Philippine Statistics Authority | Republic of the Philippines. Psa.gov.ph. <https://psa.gov.ph/>
- Philippine Statistics Authority | Republic of the Philippines. (n.d.). Psa.gov.ph. <https://psa.gov.ph/classification/psoc/technical-notes>

Assessment Rubrics:

Criteria	Description	Points	Score
DATA PRE-PROCESSING			
Introduction & Rationale	The rationale behind selecting the specific problem or objective for the project is well-explained, including the factors that influenced the decision-making process.	20	
Analytics Framework	The analytics framework guiding the achievement of the final project is well-presented. The problem or objectives are clearly defined, showing a solid understanding of the project's aim.	20	
Conclusion	Provides a clear summary of key insights gained from the dashboards and their potential impact on addressing the project objectives	10	
Subtotal		50	
DATA PRE-PROCESSING			
Data Cleaning and Preparation	Handling missing values, standardizing data formats, and removing duplicates.	20	
Feature Engineering	Creating derived attributes, transforming data for analysis. Merging or joining datasets effectively when necessary	20	
Documentation and Explanation	Clear explanation of preprocessing techniques.	10	
Subtotal		50	
DASHBOARD			
Visualization Design	Effective use of charts, clear labels, and a user-friendly layout.	20	
Insight Presentation	Highlighting key insights, meaningful comparisons, and trends.	15	
Interactivity	Inclusion of slicers, filters, and user interaction features.	10	
Aesthetic Design	Visual appeal, color scheme, and text clarity.	5	
Subtotal		50	
		150	